INTERNATIONAL MONETARY FUND

Alternative Trade Corridors of the GCC: The Case of IMEC

Mitchell W. Boice, Ken Miyajima

WP/26/237

***IMF Working Papers* describe research in progress by the author(s) and are published to elicit comments and to encourage debate.**
The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

2026
OCT

WORKING PAPER

© 2026 International Monetary Fund WP/26/237

IMF Working Paper
Middle East and Central Asia Department and Research Department

Alternative Trade Corridors of the GCC: The Case of IMEC

Prepared by Mitchell W. Boice, Ken Miyajima

Authorized for distribution by Azim Sadikov
October 2026

***IMF Working Papers* describe research in progress by the author(s) and are published to elicit comments and to encourage debate.** The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

ABSTRACT: We provide the first quantitative general-equilibrium assessment of the trade and welfare effects of the India–Middle East–Europe Economic Corridor (IMEC), a proposed rail-to-sea trade route through the Arabian Peninsula. We translate changes in transit times and transportation cost into bilateral trade costs and use them to inform a structural gravity counterfactual for over 200 countries and territories. We estimate that the declines in trade costs from IMEC would be modest, averaging 3.6 percent among affected pairs. Our results indicate that IMEC would benefit the GCC countries and Jordan most, with these countries gaining from the enhanced overland connectivity and more direct access to trade partners in Europe, North America, and North Africa. Upper- and lower-bound scenarios highlight the nonlinear role of relative rail and sea freight rates and the value of time in determining the corridor's effectiveness, while additional sensitivity exercises show that transit delays can substantially reduce route adoption and the resulting gains.

RECOMMENDED CITATION: Boice, Mitchell W. and Ken Miyajima. 2026. Alternative Trade Corridors of the GCC: The Case of IMEC. IMF Working Paper WP/26/237, Washington DC.

JEL Classification Numbers:	F14, F17, R41
Keywords:	GCC; IMEC; International trade; structural gravity; trade costs; transportation infrastructure
Author's E-Mail Address:	mboice@nd.edu, kmiyajima@imf.org

WORKING PAPERS

Alternative Trade Corridors of the GCC: The Case of IMEC

Prepared by Mitchell W. Boice, Ken Miyajima[1]

[1] The views expressed in this paper are those of the authors and do not necessarily represent the views of the International Monetary Fund (IMF), its Executive Board, or IMF management. The authors thank Robert C. Johnson, Josefine Quast, Edoardo Tolva, and Haggay Etkes for their helpful comments. We also thank seminar participants at the IMF and the Bank of Israel for their feedback. All remaining errors are our own.

1 Introduction

In recent years, shipments traversing the Middle East have faced a myriad of disruptions, including those associated with ongoing regional conflict, significantly limiting traffic through the Suez Canal, Red Sea, and Strait of Hormuz. Consequently, policymakers have become increasingly interested in the viability of alternative trade routes that could reduce dependence on the region's major maritime chokepoints. This issue is especially salient for the member states of the Gulf Cooperation Council (GCC). In March 2026, for example, the GCC Secretary General stated that "maintaining the smooth flow of land, maritime, and air transport is no longer an option, but a strategic necessity that requires additional proactive steps to. . .minimize any potential impact on trade movement and supply chains" (Gulf Cooperation Council General Secretariat, 2026).

Against this backdrop, the India-Middle East-Europe Economic Corridor (IMEC) is one prominent effort to develop an alternative trade route in the region. Launched at the G20 summit on September 9th, 2023, IMEC is a multinational initiative that envisions a new rail-based trade route across the Arabian Peninsula, featuring ship-to-rail transition hubs in the United Arab Emirates (UAE) and Israel that would not strictly require shipment via the Suez Canal, Red Sea, or Strait of Hormuz. Official projections of the corridor's benefits are sizable. Government officials from the European Union (EU) and India claim it could decrease transportation costs by 30% and transit times by 40% between their respective regions.[1] Few quantitative analyses, however, substantiate these claims, leaving the corridor's economic viability uncertain.

This paper provides the first quantitative, general equilibrium (GE) assessment of IMEC's potential impact on trade and welfare. Specifically, our analysis considers how the corridor affects over 200 countries and territories around the globe, allowing us to capture the broad trade creation and diversion forces IMEC may generate. We calculate the changes in transit times and transportation costs from the new trade route at the country-pair level and then use these estimates to inform various policy counterfactuals in a structural gravity model. Our analysis reveals a tension at the heart of IMEC that is unique to each bilateral trade partnership: the corridor effectively enables faster shipping times between countries shipping through the Middle East, but the rail-based nature of IMEC often requires a sizable increase in freight costs. Since the IMEC segment represents a different share of total shipping costs across country pairs, and because country pairs differ in their baseline reliance on Middle East shipping routes, this tradeoff creates substantial heterogeneity in the impacts of IMEC, even among its intended participants.

To place realistic bounds on the agreement's global impact, we construct three scenarios (an upper-bound, lower-bound, and central) based on existing estimates of the rail and sea freight rates, as well as the value of time (Hummels and Schaur, 2013). According to our central scenario, IMEC confers substantial reductions in trade costs for GCC countries and Jordan, where overland rail connections provide a more efficient means to transport goods than maritime alternatives. For these countries, the trade route also provides more direct east-west access to destinations in Europe, North Africa, and North America. For the subset of goods IMEC is likely to affect, trade costs between GCC countries and the European Union, for example, fall by approximately 3%. Broadly, however, IMEC does not lower trade costs for all of

[1] See, for example, European Commission (2023) and Press Information Bureau Delhi (2025).

its signatories. Though the new route reduces the distance between India and many of its European trade partners by close to 1,000 kilometers, in most cases, the higher price of shipping by rail outweighs any gains from reduced shipping times. More generally, trade costs around the world fall the most between countries for which the distance reductions under IMEC are large and there are long maritime segments, such that the higher costs of the overland leg have only a minimal effect on the average shipping cost. Accordingly, those countries that benefit the least from IMEC are typically in South America, Sub-Saharan Africa, East Asia, and the Pacific region.

More generally, our upper- and lower-bound scenarios emphasize the nonlinear nature of the rail-to-sea freight ratio in determining the overall performance of IMEC. Because countries can always fall back on existing maritime routes through the region, changes in the price of rail freight induce route switching on the extensive margin. As a result, a small change in the freight rate of rail versus sea can affect global patterns of trade diversion and creation, beyond simply changes in magnitudes. At the route level, the number of bilateral pairs for which IMEC offers a lower trade cost than the Suez Canal rises from 374 in the lower-bound scenario, to 1,974 in the central scenario, and to 3,580 in the upper-bound scenario. Thus, changes in relative freight rates affect not only the magnitude of IMEC's gains but also the set of bilateral relationships for which the corridor is economically viable. By varying freight rates and the value of time across the scenarios, we capture this sensitivity and identify benchmark cost ratios required for IMEC to generate meaningful gains.

Our approach to the broader question of economic viability is deliberately partial. Ultimately, the case for an alternative corridor depends on the benefits it generates during both periods of disruption and otherwise. We focus on the latter. If the benefits of IMEC absent disruptions to existing maritime routes are sufficiently large, then IMEC might be economically viable even before the additional benefits during times of disruption are accounted for. We therefore view our analysis as a first step towards assessing corridor viability rather than a comprehensive cost-benefit analysis, which would require us to take a stand on the probability and severity of future disruptions. Like other trade routes, IMEC itself would not be immune to disruptions arising from conflict escalation, infrastructure damage, or broader geopolitical risks. In our central specification, IMEC raises the annual GDP of the GCC countries, measured in constant 2019 prices (hereafter, "real GDP"), by roughly $8 billion. Relative to rough estimates of $20–30 billion in construction cost, these gains suggest that the corridor's benefits outside periods of disruption could be economically meaningful, although we strongly encourage a more rigorous cost-benefit analysis to determine whether they are sufficient to justify the investment.

Our paper contributes to several strands of literature. First, we contribute to the extensive literature using gravity techniques to study regional integration and trade policy (Anderson and Van Wincoop, 2003; Rose, 2004; Baier and Bergstrand, 2007; Egger and Larch, 2008; Caliendo and Parro, 2015), which is often concerned with the effects of tariff-driven trade liberalizations, such as the North American Free Trade Agreement or World Trade Organization membership. We differ from these papers in that we focus on reductions in nontariff trade barriers, namely, improvements to shipping routes connecting the regions in the agreement. Second, we contribute to the literature on trade and transportation infrastructure (Donaldson and Hornbeck, 2016; Donaldson, 2018; De Soyres, Mulabdic and Ruta, 2020; Bird, Lebrand and Venables, 2020; Allen and Arkolakis, 2022). The closest paper to ours is De Soyres et al. (2019), which quantifies the

changes in trade costs associated with the Belt and Road Initiative (BRI).

Relative to this and related work, our setting differs in two important respects. Much of the existing transportation-infrastructure literature studies investments that expand or improve existing transportation networks. IMEC instead introduces a competing intermodal route in a region already served by established maritime connections, most importantly the Suez Canal. This distinction makes route choice and the relative costs of competing transportation modes central to the corridor's effects. Moreover, while many studies emphasize the time savings generated by transportation infrastructure, we explicitly incorporate freight costs alongside time considerations. This distinction is essential in our setting because the faster rail segment of IMEC must compete directly with comparatively inexpensive maritime shipping. Finally, we contribute to the literature on the economic importance of the Suez Canal (Feyrer, 2021; Verschuur, Lumma and Hall, 2025) by evaluating the conditions under which an alternative corridor could compete with canal-based shipping.

The rest of the paper proceeds as follows. Section 2 provides an overview of the IMEC initiative and the latest details regarding its implementation. Section 3 presents our data, assumptions, and counterfactual design. Section 4 reports the resulting changes in trade costs, trade flows, and welfare, examines the sensitivity of route adoption to relative freight rates and transit delays, and discusses implications for GCC trade diversification and implementation. Finally, Section 5 concludes.

2 Background on IMEC

Before advancing to the analysis, this section provides an overview of the corridor, summarizing both the provisions described at its inception and subsequent developments. Official documentation on IMEC remains limited, and the sole official source is a memorandum of understanding (MoU) signed by the governments of India, the United States (US), Saudi Arabia, the UAE, France, Germany, Italy, and the European Union at the G20 summit on September 9, 2023. The MoU outlines broad initiatives but stops short of specifying detailed steps regarding implementation, financing, or timelines. Consequently, while the discussion in this section draws primarily from the MoU, we supplement it with news coverage and statements from policymakers where necessary to provide a more complete description of the corridor.

Taken as a whole, IMEC can be classified as a multifaceted strategy for further economic integration between Europe, the Middle East, and Southwest Asia. Its landmark feature is the introduction of an overland trade route crossing the Arabian Peninsula. Illustrated in Figure 1, IMEC would connect the Mediterranean and Arabian seas via a cross-border rail system, providing an intermodal sea-to-rail-to-sea shipping alternative to the Suez Canal. As discussed in Section 1, policymakers and government officials are optimistic about the potential cost savings from the corridor, citing a potential reduction in transportation costs of 30% and a decrease in transit times of 40%. With few quantitative analyses to support them, these figures appear to be premised on the overall decrease in distance and increase in rail speeds relative to shipping distances and times for maritime routes. We calculate that the IMEC route, for example, would cut the distance between Athens and Mumbai by nearly 1,000 km or 12% of the distance of the Suez Canal route. Moreover, trains often travel two to three times faster than sea vessels, so significant time savings are plausible for

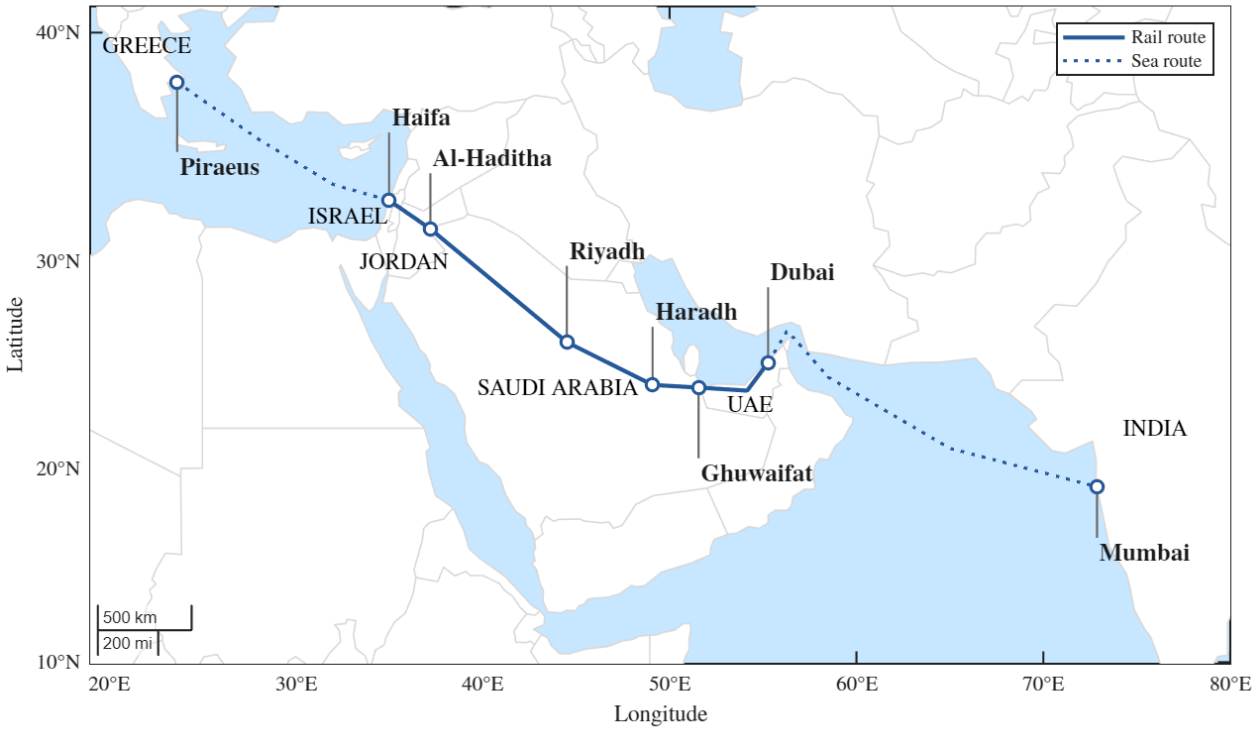


Figure 1: A map of the India–Middle East–Europe Economic Corridor (IMEC). White bubbles mark relevant nodes along the trade route, the solid blue line depicts the overland railway, and the dotted blue lines represent maritime legs.

these routes.

While not explicitly identified in official documents, Haifa, Israel and Dubai, UAE are the purported termini for the overland route (Fredman and Rozen, 2025). Notably, the proposed IMEC route predates the recent disruption to shipping through the Strait of Hormuz, which has prompted discussion of extending the eastern terminus to UAE ports outside the strait, such as Khor Fakkan or Fujairah.[2] We retain Dubai as the terminus in our analysis, though shifting to either alternative port would have little quantitative effect on our results because the corresponding change in distance is small relative to the length of the overall route. Moreover, the MoU distinguishes two subcorridors along the route: an East Corridor linking India to the Arabian Gulf and a North Corridor connecting the Arabian Gulf to Europe. However, this distinction appears largely descriptive rather than operational, with no information provided regarding logistical or organizational differences between each subcorridor.

Beyond transportation infrastructure, the MoU specifies complementary initiatives, such as the installation of a pipeline to support hydrogen exports and the laying of submarine cables to enhance energy transmission and digital connectivity, respectively. Notably, these brief references exhaust the scope of the official documentation on IMEC. The MoU concludes by calling on participating governments to reconvene within 60 days to define concrete actions and timelines. To our knowledge, progress halted less than

[2]See Chaudhury (2024), DD India (2026), and Emirates News Agency (2026) for more reports on IMEC's eastern terminus.

a month after the MoU was signed. Official discussions surrounding IMEC resumed in early 2025 with the inauguration of US President Donald Trump and a visit to India by European Commission President Ursula von der Leyen, but no further substantive details have been disclosed publicly. The lone exception to the stalled progress on IMEC has been the India-UAE Virtual Trade Corridor announced in 2024, which aims to harmonize the regulatory and customs systems of India and the UAE via a new digital interface.

Given the limited institutional clarity, our analysis focuses on IMEC's principal and most clearly articulated component: the intermodal trade route spanning the Arabian Peninsula. We operationalize the corridor around the following assumptions. First, we assume the overland segment consists of approximately 2,500 kilometers of rail, following the alignment depicted in Figure 1. From our survey of various policy reports and existing infrastructure, we find this routing to be the most plausible, as it connects many major cities and ports in a relatively direct manner. Although IMEC could complement the proposed GCC Rail System connecting all GCC countries via common rail infrastructure (Suri et al., 2024), we abstract from these connections and consider only the main Dubai-Haifa rail line. Accordingly, trade involving other GCC countries must rely on intermodal connections. For example, exports from Greece to Kuwait would first travel by sea to Haifa, by rail to Dubai, and then by sea to Kuwait. Our model does not allow for goods to travel directly from Haifa to Kuwait by rail.

Second, despite the critical role of Israel and Jordan in the overland route, neither country has issued formal public commitments regarding participation. Thus, to evaluate the potential of the corridor, we take both countries' involvement as given. Last, while the principal aim of the agreement is to foster trade among India, the GCC, and Europe, the MoU does not preclude other regions from using the overland route. Thus, our model permits all countries to route trade through IMEC, allowing us to assess the corridor's global trade and welfare implications rather than limiting attention to signatory economies.

3 Gravity Model

This section details the construction of our gravity counterfactual. We begin by broadly discussing our approach. Then, we report our data sources and estimates of key structural parameters. Finally, we present our methodology for converting changes in time and freight costs into counterfactual bilateral trade costs.

3.1 Conceptual Framework

To discipline our analysis, we begin by laying out several modeling assumptions that warrant discussion upfront. First, our exercise views IMEC through the lens of a structural gravity model. Although the gravity framework is the gold standard for quantitative trade policy analysis, it is mildly restrictive in our setting. In particular, conventional gravity models excel at quantifying how bilateral trade costs shape bilateral trade flows, but they typically abstract from the specific trade routes through which goods move from source to destination. In the case of IMEC, this distinction is particularly important, as its economic effects depend not only on changes in trade costs but also on whether shippers substitute away from the incumbent route through the Suez Canal. A burgeoning literature examines endogenous trade costs to address precisely this issue: agents optimally choose shipping routes, and trade costs emerge as an equilibrium function of route

traffic.[3] Though such frameworks are also suited to the study of IMEC, they are often computationally intensive and require granular, proprietary shipping data. We therefore retain the tractability of the gravity model, but augment the framework by introducing a simple route selection mechanism.[4]

Specifically, we allow for shipments transiting the Middle East to choose between the Suez Canal and IMEC routes. We assume that these routes are perfectly substitutable, such that goods initially routed through the Suez Canal deterministically switch to IMEC whenever the latter offers a lower route-specific trade cost. This extension allows the model to capture an extensive margin of route adoption that is absent from a conventional gravity framework. Changes in the relative cost of shipping via the two routes not only affect the magnitude of bilateral trade costs, but also whether a given country pair uses IMEC at all. At the same time, our treatment of route selection remains deliberately parsimonious. We do not model endogenous route choice across the full global transportation network and therefore cannot capture substitution from non-Suez alternatives, such as the Cape of Good Hope. Nor do we allow route costs to respond endogenously to congestion, capacity constraints, or traffic volumes. Our results should therefore be interpreted as capturing route substitution between IMEC and the Suez Canal rather than a full equilibrium model of global shipping routes.

Finally, recall that the IMEC route requires goods to be shipped from sea to rail and back to sea. Motivated by the transportation and logistics literature, we assume that only containerized goods (i.e., those that can travel in standardized twenty- or forty-foot shipping containers) are affected by the agreement. Because repackaging goods at each modal shift would incur substantial handling and switching costs, recent studies contend that containerization is the most efficient means of intermodal transport (Chao, Yu and Hsieh, 2018; Abu-Aisha, Audy and Ouhimmou, 2024). In contrast, non-containerized goods, such as bulk commodities and crude oil, often require specialized vessels, which substantially limits their suitability for intermodal shipment (Hummels, 2007). We follow Bernhofen, El-Sahli and Kneller (2016) in identifying a containerizable goods sector from disaggregated trade flows. Because the gravity model is additive, we conduct our analysis on this sector alone, and all non-containerized goods shipping via the Suez Canal are assumed to be unaffected by IMEC. We also omit any cross-sector effects from our analysis and leave any multisector extensions to future work.[5]

3.2 Data

Structural gravity models require a balanced panel of bilateral trade flows, including both international and domestic observation (Yotov et al., 2016). Accordingly, we use the International Trade and Production Database for Simulation (ITPD-S) for our counterfactual analysis (Borchert et al., 2024). The ITPD-S uses

[3] See, for example, Baccelli and Morino (2020) and Ganapati, Wong and Ziv (2024)

[4] Our choice of model is also consistent with related research. For example, De Soyres, Mulabdic and Ruta (2020) analyzes similar infrastructure developments related to the BRI within a gravity framework.

[5] Explicitly, this means that our analysis abstracts from upper-level substitution effects. As trade costs fall for containerizables in a given country, real output in that sector may expand. In general equilibrium, this would generate income effects that could increase demand for both containerizable and non-containerizable goods, as well as substitution effects across sectors. By focusing solely on the containerizable sector, we ignore these cross-sector reallocations and treat non-containerizable trade as unaffected by IMEC.

raw trade flows from the United Nations Commodity Trade Statistics Database (COMTRADE) but makes two important adjustments. First, the ITPD-S includes domestic trade flows, which are taken from various administrative data sources depending on the type of good.[6] Second, the ITPD-S estimates missing trade flows to create a balanced panel of both international and domestic trade data for 170 industries in 265 countries for the years 1986–2019.

Since the ITPD-S uses imputed values to form a balanced panel, however, it is not suited for gravity estimation. Therefore, we use the International Trade and Production Database for Estimation (ITPD-E) to calibrate the trade elasticity for our model (Borchert et al., 2021; Larch, Shikher and Yotov, 2025). The ITPD-E and ITPD-S were designed to be used in tandem, with observations in both datasets being identical wherever possible. We merge these data with the Dynamic Gravity Dataset made available by Gurevich and Herman (2018), which contains a broad set of macroeconomic indicators, geographic and cultural characteristics, and trade facilitation measures. We also include bilateral tariffs from the CEPII's Market Access Map-HS6 (MAcMap) dataset (Guimbard et al., 2012). MAcMap provides applied tariff duties for 152 importing countries and 189 exporting countries for 2001–2019 at the six-digit product level (Harmonized System). We construct simple and trade-weighted averages to aggregate these data to sectors consistent with the ITPD-E.

We also use a number of sources relevant to our objective of evaluating the savings of IMEC over Suez transit. First, following Bernhofen, El-Sahli and Kneller (2016), we focus on the containerizable goods sector. In 1968, the Verband Deutscher Ingenieure (German Association of Engineers) classified SITC four-digit codes by their containerizability as follows: containerizable products, products with limited containerizability, and noncontainerizable products. We focus on the 712 fully containerizable products, and our results are robust to including those products with limited containerizability. Appendix A describes how we aggregate product-level containerizability into the ITPD-E and ITPD-S nomenclature and provides a list of containerizable sectors for our analysis. Of 170 sectors, we ultimately label 92 as containerizable, the majority of which are manufacturing industries.

Second, because our paper focuses on maritime shipping, we employ bilateral sea distances from the CERDI-SeaDistance database (Bertoli, Goujon and Santoni, 2016). Bilateral maritime distances are computed as the shortest existing sea route between two countries, where the relevant ports are those with the highest number of shipping lines. All landlocked countries are associated with the foreign port with the shortest road distance to the capital city. These data are available for 227 countries and territories, and any missing sea distances are filled in with the distance variable provided in the Dynamic Gravity Dataset. Including these data in our analysis allows us to more accurately calculate the baseline trade costs associated with shipping via the Suez Canal. Relatedly, this paper abstracts from the impact of lost toll and fee revenues on the host countries stemming from route switching.

Finally, we leverage the data on maritime shipping through global chokepoints provided by Verschuur, Lumma and Hall (2025), who in turn build on IMF Portwatch data (Verschuur et al., 2025). Specifically, we calculate the share of trade that transits the Suez Canal for each country pair. Doing so allows us to

[6]For example, data on domestic agricultural production come from the Food and Agriculture Organization of the United Nations Statistics Division, while the data on the domestic production of manufacturing goods are sourced from the United Nations Industrial Statistics database.

gauge a country's exposure to the IMEC route, since countries that already do not use the Suez Canal are unlikely to use IMEC when it opens. Importantly, the data come from 2022, before disruptions in the Red Sea substantially reduced trade volumes through the Suez Canal. Combining all of our data sources, we are left with 189 countries and territories from 2001 to 2019 for estimation and 226 in the year 2019 for simulation.

3.3 Estimating the Trade Elasticity and Baseline Trade Costs

This section follows the model and procedure for estimating parameters outlined by Yotov et al. (2016). Because the model is entirely standard, we skip the derivations and summarize the GE system required for our work.[7][8] Since the gravity model holds for each sector and in each time period, we omit sector- and time-specific notation and add these back in only when they are necessary. The structural gravity model, which describes global trade flows in containerized goods, appears in *source–destination* format as follows

$$X_{ij} = \frac{Y_i E_j}{Y} \left(\frac{t_{ij}}{\Pi_i P_j} \right)^{1-\sigma}, \tag{1}$$

$$\Pi_i^{1-\sigma} = \sum_j \left(\frac{t_{ij}}{P_j} \right)^{1-\sigma} \frac{E_j}{Y}, \tag{2}$$

$$P_j^{1-\sigma} = \sum_i \left(\frac{t_{ij}}{\Pi_i} \right)^{1-\sigma} \frac{Y_i}{Y}, \tag{3}$$

$$p_i = \left(\frac{Y_i}{Y} \right)^{\frac{1}{1-\sigma}} \frac{1}{\Pi_i}, \tag{4}$$

$$E_i = \varphi_i Y_i. \tag{5}$$

In the above system, X_{ij} denotes the containerized imports of country j from source country i, which are subject to the usual iceberg trade costs, $t_{ij} \geq 1$. Π_i denotes the outward multilateral resistance, which represents the costs of selling one unit abroad, as if country i were shipping to a unified global market. Similarly, P_j represents the inward multilateral resistance, which captures the costs of buying one unit from a unified global market. p_i, E_i, and Y_i represent factory-gate prices, total expenditures, and the value of total containerized output in country i, respectively. Y denotes the value of world output, and $\sigma > 1$ is the elasticity of substitution for containerized goods across countries. Last, φ_i defines the relationship between aggregate output and expenditure such that $\varphi_i > 1$ implies a trade deficit and $\varphi_i < 1$ implies a trade surplus.

Most objects in the model can be inferred directly from data. For example, φ_i is chosen so that model trade imbalances match those observed in the data. However, we need to estimate the trade elasticity for containerized goods, σ, and our baseline set of trade costs, t_{ij}. To estimate the trade elasticity, we specify

[7] Our model is GE conditional on initial endowments, as is standard, and, in our case, holding trade and output in all noncontainerizable sectors fixed.

[8] See Eaton and Kortum (2002) and Anderson and Van Wincoop (2003), for microfoundations of the gravity framework.

Table 1: Estimating the Trade Elasticity of Containerized Goods

Variable	(1)	(2)	(3)	(4)
τ_{ijt}	-10.17***	-11.37***	-8.80***	-5.48***
	(1.12)	(1.01)	(1.50)	(0.34)
$\ln(dist_{ij})$	-0.49***	-0.48***	-0.51***	-1.39***
	(0.05)	(0.05)	(0.07)	(0.02)
$contiguous_{ij}$	0.52***	0.50***	0.45***	0.71***
	(0.14)	(0.12)	(0.15)	(0.12)
$commoncolonizer_{ij}$	0.57***	0.67***	0.05	0.77***
	(0.17)	(0.17)	(0.28)	(0.05)
$language_{ij}$	0.22**	0.22**	0.14	0.84***
	(0.11)	(0.10)	(0.12)	(0.04)
FTA_{ij}	0.20**	0.00	0.36***	0.48***
	(0.09)	(0.09)	(0.12)	(0.04)
$international_{ij}$	-2.15***	-1.84***	-2.17***	-2.24***
	(0.18)	(0.19)	(0.26)	(0.29)
PPML	x	x	x	
Import-Weighted Tariff	x		x	x
Importer-Time FEs	x	x		x
Exporter-Time FEs	x	x		x
Observations	188,724	188,724	132,933	141,792
R-Squared	0.98	0.98	0.94	0.77

Standard errors clustered by country pairs.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the standard gravity equation from Yotov et al. (2016):

$$X_{ijt} = \exp\left(\pi_{it} + \chi_{jt} + \beta_0 + \beta_1 \ln dist_{ij} + \beta_2 \tau_{ijt} + \beta_3 G_{ij}\right) \times \varepsilon_{ijt}, \tag{6}$$

Here, X_{ijt} denotes the observed containerized trade flow from source i to destination j at time t. π_{it} and χ_{jt} are exporter-time and importer-time fixed effects, respectively, which control for the multilateral resistance terms outlined in Equations (2) and (3), respectively. $\ln dist_{ij}$ is the log of the bilateral distance between country pairs, τ_{ijt} is the trade-weighted average tariff, and G_{ij} is a vector of common gravity variables, such as the sharing of a common border, language, colonizer, etc. Because tariffs directly shift trade costs, we recover the trade elasticity from the estimate of β_2. We estimate Equation (6) with the Poisson pseudo-maximum likelihood (PPML) estimator at three-year intervals from 2001-2019 using the data from the ITPD-E, Dynamic Gravity Dataset, and MAcMap.

Table 1 presents our results across four specifications commonly found in the gravity literature. The estimates in Column 1, our main specification, show the trade elasticity of containerizable goods to be 10.2, well

in line with similar estimates across the literature (Eaton and Kortum, 2002; Anderson and Van Wincoop, 2003; Broda, Greenfield and Weinstein, 2006). The results in Column 2 correspond to the same equation as in Column 1, except that the specification uses a simple average rather than a weighted average tariff measure. In Column 3, we remove the importer- and exporter-time fixed effects, proxying them with additional country-level information such as GDP, population, and the costs of doing business. The differences in the trade elasticity across Columns 1, 2, and 3 are not statistically significant from zero. Column 4 reports ordinary least squares (OLS) estimates of a similar gravity specification, which produces a notably lower trade elasticity. Because OLS requires logged trade flows, many observations for high-tariff country pairs are omitted because the observed trade flow is zero, which biases the estimate downward.

Since our specification in Column 1 aligns most closely with conventions in the literature, we set the trade elasticity for our counterfactual exercise to 10.2. To further validate our estimate, we benchmark our value against the trade elasticities estimated by Fontagné, Guimbard and Orefice (2022), which are reported at the six-digit product level. Restricting their sample to products that are containerizable, we obtain a weighted (unweighted) average trade elasticity of 10.9 (9.2), which closely matches our central estimate.

To obtain our baseline trade costs for the counterfactual, we re-estimate Equation (6) without including bilateral tariffs, which maintains the flexible form of trade costs without sacrificing observations due to missing data. We use the PPML estimator and data from the ITPD-S for 2019, the most recent year of data available at time of writing. We recover our baseline trade costs from the estimated coefficients on the gravity variables.[9]

3.4 Structural Counterfactual

To estimate the impact of the IMEC agreement, we construct a counterfactual set of bilateral trade costs and feed these through a structural gravity model, holding all other parameters fixed. The credibility of our results depends crucially on how well our counterfactual environment reflects the implementation of the IMEC agreement. As discussed in Section 2, IMEC is primarily a logistical initiative aimed at establishing an alternative to the Suez Canal trade route rather than a policy intervention designed to reduce either tariff or nontariff barriers. Our analysis, therefore, concentrates on quantifying the reduction in shipping-related trade costs—namely time and transportation costs. We present the derivation of our counterfactual trade costs in reverse order, starting from the changes in times and transportation cost at the route level before showing how these aggregate to country-level bilateral trade costs.

A key challenge we face in assessing both time and transportation costs is capturing the total distance traveled between two countries along a given route, rather than focusing narrowly on the length of the Suez Canal or IMEC corridor itself. This distinction is quantitatively important in that country pairs for which the Suez Canal route segment corresponds to a large share of the total distance traveled (e.g., Greece–India) should see larger changes in trade costs than pairs for which the canal segment represents a lesser share (e.g., United States–India). To address this issue, we rely on bilateral sea distances rather than great-circle distances. Specifically, we interpret the raw bilateral distances from the CERDI-Seadistance Database as the

[9]See Yotov et al. (2016) for more details on calibrating the baseline parameters. Specifically, we follow the estimation procedure described in Section B of Chapter 2.

length required for two countries to trade via the Suez Canal.[10] To construct the IMEC distances, we first calculate the maritime distance from the exporting country to the nearest IMEC terminus—Haifa, Israel or Dubai, UAE. We then add the estimated 2,500 kilometers of rail connecting Haifa to Dubai and the maritime distance from the opposite IMEC terminus to the importing country.[11]

With these distances accounted for, we compute the transportation costs associated with each route. For shipments routed through the Suez Canal, the goods travel exclusively by sea, implying that transportation costs can be written as

$$C_{ij}^{Suez} = r^{sea} \times seadist_{ij}, \tag{7}$$

where r^{sea} denotes the maritime freight rate. In contrast, transportation costs along the IMEC route reflect a combination of maritime and rail segments, appearing as

$$\begin{aligned} C_{ij}^{IMEC} = & \mathbf{1}\left\{seadist_{i,ISR} < seadist_{i,UAE}\right\} \times r^{sea} \times (seadist_{i,ISR} + seadist_{UAE,j}) \\ & + \mathbf{1}\left\{seadist_{i,ISR} > seadist_{i,UAE}\right\} \times r^{sea} \times (seadist_{i,UAE} + seadist_{ISR,j}) \\ & + r^{rail} \times 2500, \end{aligned} \tag{8}$$

where the indicator functions select the IMEC terminus closest to the exporting country and r^{rail} denotes the railway freight rate. This formulation captures differences in the transportation costs arising from the use of maritime versus rail transport. We abstract, however, from port and canal fees on both routes, primarily because we lack reliable data. At time of writing, IMEC remains in an early planning stage; therefore, no reliable information is available regarding fees that may be levied on shipments traversing the corridor. Although the Suez Canal Authority publicly reports toll schedules, these fees vary substantially by vessel type, size, and net tonnage, making it difficult to construct a representative fee applicable across country pairs. More broadly, the opening of the IMEC route is likely to affect pricing by the Suez Canal Authority, rendering current tolls an unreliable benchmark. In any case, since port or canal fees likely constitute a negligible share of a good's value, we believe this omission does not meaningfully affect our results or conclusions.[12]

We compute time costs using route-specific distances, the average speed by transportation mode, and the average processing time for ships spent in port. Because rail transport is substantially faster than maritime shipping, time savings are the primary means through which IMEC would lower trade costs. Following De Soyres et al. (2019), we assume that maritime shipping travels at 25 kilometers per hour and that rail freight travels at 75 kilometers per hour. Furthermore, we use estimates of port processing time from Slack et al. (2017), which are displayed in Table 2. With these in hand, we denote the time required to traverse the

[10] Although the shortest distance between two countries does not always use the Suez Canal, the subsequent construction of trade costs will correct for any distortions this interpretation may introduce.

[11] While this operation accurately calculates the total distance required to use the IMEC route between two countries, we make adjustments for both Jordan and Saudi Arabia, who do not need to traverse the entire length of the IMEC corridor when trading. In these cases, we sum the maritime distance from each partner country to the closest IMEC terminus with the distance from that IMEC terminus to either Jordan or Saudi Arabia.

[12] Per transit, an average ship passing through the Suez Canal faces an estimated average toll of $465,000 (Chu and Ledermann, 2017), and the average value of trade per twenty-foot equivalent unit (TEU) in maritime trade is $54,500.

Suez Canal in hours as

$$h_{ij}^{Suez} = \frac{seadist_{ij}}{v^{sea}}, \tag{9}$$

where v^{sea} is the maritime travel speed. We calculate the time to traverse the IMEC route as

$$\begin{aligned} h_{ij}^{IMEC} = & \mathbf{1}\left\{seadist_{i,ISR} < seadist_{i,UAE}\right\}\left(\frac{seadist_{i,ISR} + seadist_{UAE,j}}{v^{sea}}\right) \\ & + \mathbf{1}\left\{seadist_{i,ISR} > seadist_{i,UAE}\right\}\left(\frac{seadist_{i,UAE} + seadist_{ISR,j}}{v^{sea}}\right) \\ & + \frac{2500}{v^{rail}} + porttime_{ISR} + porttime_{UAE}, \end{aligned} \tag{10}$$

where v^{rail} is the railroad travel speed and $porttime_{ISR}$ and $porttime_{UAE}$ are the port processing times. To convert transit times into an iceberg trade cost form that we can use in our model, we take estimates of the value of time from Hummels and Schaur (2013). The time costs of each route $u \in \{\text{Suez, IMEC}\}$ are given by

$$H_{ij}^{u} = \exp\left(\frac{\theta \times h_{ij}^{u}}{24}\right). \tag{11}$$

In the above expression, θ denotes the time valuation parameter, which captures the effect of a one-day increase in delivery times on trade costs. Equation (11) converts the total days in transit from each route into a tariff-equivalent form. For further discussion on the time cost of trade, we direct the reader to Hummels and Schaur (2013).

Table 2: Average Port Processing Time by Region

Region	Mean Ship Time in Port (hours)
Central and South America	23.5
South and Southeast Asia	26.5
East and North Asia	17.2
Gulf and Red Sea	26.8
Mediterranean	20.3
Northern Europe	29.5
North America – East Coast	21.1
North America – West Coast	46.2
South Africa	64.6

Note: Table reports mean time (in hours) that ships spend in port for each region. Estimates come from Slack et al. (2017).

To aggregate the time and transportation components, we adopt a log-linear (multiplicative) specification for bilateral trade costs. While this formulation is the most common in the gravity literature, it is not the only possible choice. For instance, De Soyres, Mulabdic and Ruta (2020) opt instead for an additive trade cost specification. Since at least Anderson and van Wincoop (2004), the appropriate form of the trade cost function has been a debated topic in the gravity literature, but its implementation in practice has ultimately

been determined by whichever form is best suited for the question at hand.[13] We assume a multiplicative formulation because it allows us to express proportional changes in aggregate trade costs using only the components directly affected by IMEC. By contrast, an additive specification of trade costs would require us to measure other elements of trade costs that are orthogonal to the focus of our analysis. Moreover, aggregate trade costs are often decomposed multiplicatively into transportation costs and tariffs (Egger et al., 2021), which aligns naturally with our separation of transportation costs and tariff-equivalent time costs.

More specifically, we assume that the total trade cost along each route u is given by

$$t_{ij}^{u} = C_{ij}^{u} H_{ij}^{u} R_{ij}^{u}, \tag{12}$$

where R_{ij}^{u} represents all other trade costs that remain unaffected by the IMEC agreement, such as tariffs and regulatory barriers, and therefore satisfies $R_{ij}^{Suez} = R_{ij}^{IMEC}$. Under our assumption that the Suez Canal and IMEC are perfect substitutes, shipments initially routed through the Suez will adopt IMEC whenever the latter offers a lower total trade cost. Let $\hat{t}_{ij}^{ME}$ denote the proportional change in bilateral trade costs for goods traversing the Middle East. Then,

$$\hat{t}_{ij}^{ME} = \min\left\{1, \frac{t_{ij}^{IMEC}}{t_{ij}^{Suez}}\right\} = \min\left\{1, \frac{C_{ij}^{IMEC} H_{ij}^{IMEC}}{C_{ij}^{Suez} H_{ij}^{Suez}}\right\}, \tag{13}$$

which accounts for the trade-off between faster transit times and higher transportation costs associated with rail relative to maritime shipping. Importantly, changes in time costs and transportation cost do not (and need not) have symmetric effects on aggregate trade costs. As shown in Equation (11), a higher value of θ magnifies changes in shipping times into larger changes in total trade costs. θ therefore governs the relative importance of time costs in the aggregate and allows us to construct upper and lower bounds for the impact of IMEC.

To derive the change in aggregate trade costs, we first note that there are often multiple shipping routes between any origin and destination, even for containerized trade. As a result, a reduction in trade costs along any one route should translate to a smaller reduction in aggregate trade costs since the latter reflect shipments using multiple routes. To capture this feature, we assume that aggregate bilateral trade costs, t_{ij}, are a geometric weighted average of the trade costs incurred along routes transiting the Middle East and along alternative routes. In particular,

$$t_{ij} = \left(t_{ij}^{ME}\right)^{s_{ij}} \left(t_{ij}^{alt}\right)^{1-s_{ij}}, \tag{14}$$

where t_{ij}^{ME} and t_{ij}^{alt} denote the trade costs associated with Middle East and alternative routes, respectively, and s_{ij} is the share of bilateral trade that flows through the Middle East, which we measure as trade through the Suez Canal. Equation (14) is consistent with existing literature that decomposes aggregate trade costs

[13] For example, De Soyres, Mulabdic and Ruta (2020) adopt an additive specification because it aligns with the data and modeling framework of the Global Trade Analysis Project (GTAP), which provides additive decompositions of bilateral trade costs derived from a computable general equilibrium model. These data are proprietary and therefore not used in our analysis.

Table 3: External Parameter Calibration

Description	Parameter	Central	Upper	Lower	Source
Value of Time	θ	0.021	0.021	0.006	Hummels and Schaur (2013)
Maritime Speed	v^{sea}	25 km/h	25 km/h	25 km/h	De Soyres et al. (2019)
Rail Speed	v^{rail}	75 km/hr	75 km/h	75 km/h	De Soyres et al. (2019)
Maritime Freight Rate	r^{sea}	1	1	1	Normalization
Rail Freight Rate	r^{rail}	1.93	1.55	2.49	Many[a]

[a] Central: United Nations Conference on Trade and Development (2025*b*). Upper: Hoffmeister et al. (2022) and John A. Volpe National Transportation Systems Center (2021). Lower: United Nations Conference on Trade and Development (2025*b*).

by mode of transportation. For example, Feyrer (2019) models aggregate trade costs as a similar average of air and sea costs, estimating mode-specific weights via a gravity specification. Our approach differs in that we fix these weights to the empirically observed share of bilateral trade transiting the Suez Canal, instead of inferring these weights through regression.[14]

Because the opening of the IMEC corridor will affect only shipments that would otherwise transit the Suez Canal, the proportional change in aggregate bilateral trade costs simplifies to

$$\hat{t}_{ij} = \left(\hat{t}_{ij}^{ME}\right)^{s_{ij}}, \tag{15}$$

which has an intuitive interpretation. Suppose IMEC uniformly lowers trade costs for Middle Eastern routes by 30%, such that $\hat{t}_{ij}^{ME} = 0.7$. For trade between the United States and Mexico, Equation (15) implies no change in bilateral trade costs since $s_{USA,MEX} = 0$. By contrast, trade between the United States and the UAE should experience a much larger reduction in trade costs, as approximately 60% of the containerized trade between them flows through the Suez Canal. This formulation reflects the notion that country pairs with little or no reliance on the Suez Canal are unlikely to benefit from IMEC, while the largest reductions in trade costs should accrue to country pairs with the highest dependence on the canal.

Last, we describe our choices for externally calibrated parameters, as well as the solution method. We solve our counterfactual under three scenarios. First, we consider a central scenario, in which the parameters take on their most plausible values based on our survey of the literature. Then, we consider upper- and lower-bound scenarios, where we select the parameter values from the literature that are most and least favorable to the success of the IMEC corridor. These scenarios allow us to characterize the most likely outcome of IMEC and the range of possible effects. A summary of our parameter choices and scenarios is presented in Table 3.

Our parameter values for modal speed and the value of time reflect commonly selected values in related literature. We take estimates of the value of time from Hummels and Schaur (2013). In our central scenario, we choose the value corresponding to the upper bound to stay consistent with studies similar to ours (De Soyres, Mulabdic and Ruta, 2020). Similarly, we assume the same values for maritime and rail speed as

[14] More specifically, Feyrer (2019) assumes a log-linear form for trade costs: $\ln(t_{ijt}) = \beta_{sea,t} \ln(seadist_{ij}) + \beta_{air,t} \ln(airdist_{ij}) + \beta X_{ij}$, where $\beta_{sea,t}$ and $\beta_{air,t}$ are time varying weights on sea and air transport, respectively, and distances proxy for route-specific trade costs. Our setting abstracts from X_{ij} as our objective is simulation rather than estimation.

De Soyres et al. (2019). Modal freight rates, however, display considerably more variation in the literature. Given Equations (7) and (8), our counterfactual analysis requires only the relative freight rate of rail versus maritime shipping. We therefore normalize the maritime freight rate to unity and set the rail freight rate to reflect freight ratios observed in the data.

In our central parameterization, we set the rail freight rates to be 93% higher than maritime rates. This value corresponds to the average relative ad valorem freight rate observed in the United Nations Trade-and-Transport Dataset (United Nations Conference on Trade and Development, 2025*b*) for the four most recent years of data (2018–2021). We use these years as opposed to the entire sample because these observations better reflect the recent narrowing of the gap between rail and sea freight, which is driven primarily by increases in maritime shipping costs (United Nations Conference on Trade and Development, 2025*a*). Our upper-bound scenario corresponds to the smallest margin observed in the Trade-and-Transport Dataset and also matches relative freight prices in the US Department of Transportation's National Transportation Statistics Database (NTSD) (John A. Volpe National Transportation Systems Center, 2021). Finally, our lower-bound scenario sets rail freight rates at 2.49 times as expensive as sea, which corresponds to the largest freight ratio observed for containerizable goods between 2016 and 2021 in the Trade-and-Transport Dataset.

To simulate each of our counterfactual scenarios, we follow the algorithm prescribed by Yotov et al. (2016). A sketch of the procedure is provided here, and we refer the interested reader to Chapter 2 of the original work. From Equation (15), we construct our counterfactual set of trade costs as follows:

$$t'_{ij} = t_{ij} \times \hat{t}_{ij}. \tag{16}$$

Next, we reestimate the exporter and importer fixed effects in Equation (6) in a restricted regression, holding the pair fixed effects at their values implied by our counterfactual trade costs. With the recovered outward and inward multilateral resistance terms, we solve for all other objects of the model and predict trade flows. We continue reestimating fixed effects from the new set of predicted trade flows and constructing model objects until the change in factory-gate prices from iteration to iteration is sufficiently small.

4 Results

This section presents our estimates for the impact of IMEC on global trade flows and welfare. We begin by discussing how the corridor affects trade costs, which establishes the intuition for our main results in a partial equilibrium (PE) framework. Examining trade costs also serves as a validation exercise for our approach, since our counterfactual should reflect the geographical features of shipping through the Middle East. We then examine implications for containerized and aggregate trade and containerized real GDP—a sufficient statistic for welfare in our model. Next, we assess the sensitivity of these results to relative rail and sea freight costs and transit delays. We also establish benchmarks for the cost efficiency required for IMEC to remain competitive with existing maritime routes. Finally, we outline various operational considerations and limitations of our analysis. Because presenting the full results for over 200 countries would be cumbersome, we focus on summary tables and figures that highlight the core findings of our

Table 4: Decreases in Containerized Trade Costs (%) by Scenario

Sample	Scenario	Average	Std Dev	Min	Max
World	Lower	0.01	0.30	0.00	33.72
	Central	0.08	0.72	0.00	37.49
	Upper	0.15	1.24	0.00	40.59
IMEC Core	Lower	0.15	1.15	0.00	33.72
	Central	1.03	2.50	0.00	37.49
	Upper	1.96	4.21	0.00	40.59
Non-IMEC Core	Lower	0.00	0.00	0.00	0.00
	Central	0.00	0.11	0.00	7.26
	Upper	0.01	0.24	0.00	16.33

Note: Countries flagged "IMEC Core" include Jordan, Israel, and the GCC countries. Summary statistics are unweighted and reported as percentage decreases from baseline trade costs. Trade-weighted summary statistics yield the same relative ordering with smaller magnitudes.

analysis. Appendix B reports additional results, including comprehensive country-level outcomes across each scenario in both dollar values and percentage changes.

4.1 Trade Costs

Table 4 reports summary statistics for the decreases in bilateral trade costs across each scenario, measured as a percentage of their baseline values. Across all country pairs, the average decrease in trade costs in the central scenario is 0.08%, though this average masks considerable heterogeneity. Nearly 96% of country pairs in our sample feature no change in trade costs as a result of IMEC.[15] Among bilateral pairs experiencing a nonzero reduction, which involve 150 countries, trade costs fall by approximately 3.6% on average. Across all scenarios, the corridor confers the largest reductions to the GCC countries and Jordan, where IMEC provides a substantially more direct alternative to the Suez Canal.

Disaggregating by subsample reveals the localized nature of IMEC's trade cost effects. Country pairs in which at least one partner is Israel, Jordan, or a GCC country (hereafter referred to as the IMEC core) account for the vast majority of the global decline in trade costs. In the central scenario, the average reduction for these pairs is 1%, more than two orders of magnitude larger than the corresponding reduction for pairs involving no IMEC-core country. The declines for these pairs are also considerably more dispersed, reflecting the varying exposure of country pairs to the corridor. Country pairs involving no IMEC-core country generally experience negligible reductions in trade costs. The few exceptions, such as Cyprus, Egypt, and Iran, are located in close geographic proximity to the corridor.

[15]Based on the data from Verschuur, Lumma and Hall (2025), we find this fraction realistic. Only 15% of bilateral pairs use the Suez Canal to transport containerized goods, and for many countries using the Suez Canal (e.g. countries along the eastern coast of Africa), trade would nearly or entirely traverse the canal before reaching the nearest IMEC terminus.

Our upper- and lower-bound scenarios exhibit similar geographic patterns but differ substantially in magnitude, highlighting the sensitivity of IMEC's trade-cost effects to assumptions governing rail freight costs and the value of time savings. In the upper-bound scenario, where rail is assumed to be only 50% more expensive than sea rather than 90%, the average reduction in global containerized trade costs nearly doubles relative to the central scenario and rises to approximately 2% for IMEC-core country pairs. By contrast, the lower-bound scenario combines higher relative rail freight costs with a lower value of time, sharply limiting the corridor's competitiveness. These comparisons emphasize that reductions in geographic distance alone are insufficient to ensure IMEC's viability. Rather, the corridor must also preserve a sufficiently large time advantage relative to its higher rail freight costs.

The tension between reduced distance and increased modal costs unevenly shapes IMEC's effects across its intended trade partners. Table 5 presents the simple average percentage reduction in trade costs between IMEC regions, with Israel, Jordan, and the United States also included given their significance in the agreement. The GCC countries experience large, widespread reductions in trade costs: their costs of trade with the European Union and United States fall, on average, by approximately 3% and 4%, respectively. These declines are largely the result of the reduction in the effective distance between the GCC countries and their trade partners. For example, IMEC shrinks the distance between the UAE and Greece by almost 40%. Trade costs also fall substantially within the Arabian Peninsula, reflecting improved overland connectivity between the GCC, Israel, and Jordan.

While IMEC also reduces the distance between India and many of its trading partners in all of our scenarios, these gains are frequently overshadowed by the higher price of rail freight. As a result, India experiences a negligible average reduction in trade costs with the European Union. For example, although the new corridor shortens the distance between India and Greece by 12%, shipping by rail increases freight costs by 17%, completely negating any gains from faster shipping times. India's gains instead materialize with more distant trade partners, such as the United States, where the rail segment represents a smaller share of total freight costs and the time savings from bypassing the Red Sea and Suez Canal become worthwhile.

For India–GCC trade, the limited impact of IMEC reflects a different mechanism. The corridor's primary strength lies in facilitating overland transit across the Arabian Peninsula. However, GCC countries already have substantial port infrastructure providing direct maritime access to India, reducing the need to transport goods across the peninsula. As a result, IMEC delivers almost no reduction in trade costs for GCC exports to India. For GCC imports, we observe a similar pattern, with five GCC countries experiencing no decline. Saudi Arabia drives the modest reduction observed in India–GCC trade costs, as a sizable share of its imports from India enters via the Red Sea. More generally, the muted impact on India–GCC trade reflects a prominent pattern among country pairs in our counterfactual: IMEC is most valuable for shipments that would otherwise need to circumvent the bulk of the Arabian Peninsula. Conversely, the corridor offers little advantage for countries located on the same side of the peninsula, since existing maritime routes already provide sufficiently direct connections.

Visualizing these patterns at the global level, Figure 2 depicts the simple average reductions in trade costs under the central scenario, with darker shades of blue indicating larger effects. The GCC countries and Jordan experience the largest average declines in trade costs. For these countries, the corridor provides a more direct east–west shipping route toward the Mediterranean Sea, replacing longer passages through the

Table 5: Average Change in Containerized Trade Costs for IMEC Participants (%)

Exporter / Importer	EU	GCC	IND	ISR	JOR	USA
EU	0.0	−3.0	0.0	0.0	−3.2	0.0
GCC	−2.6	−3.0	−0.3	−3.6	−5.9	−4.0
IND	0.0	−1.7	0.0	0.0	0.0	−2.2
ISR	0.0	−4.7	0.0	0.0	−1.6	0.0
JOR	−3.2	−7.2	0.0	−4.8	0.0	−0.7
USA	0.0	−3.4	−1.2	0.0	−0.6	0.0

Note: Table reports simple average percentage changes in trade costs between the listed regions. Negative values indicate reductions in trade costs.

Red Sea and Suez Canal. Countries in North America, Europe, and North Africa receive small reductions in trade costs, on average, primarily through lower costs of trade with GCC countries. In contrast, IMEC does not meaningfully alter shipping routes for much of South America, Sub-Saharan Africa, and Southeast Asia. In these regions, either few shipments require transit through the Middle East or the time savings from IMEC prove insufficient to offset the higher costs of rail transport.

The magnitudes of our counterfactual trade costs differ from those in existing studies, but in ways that are consistent with the narrower geographic scope of IMEC. The closest comparison to our counterfactual comes from De Soyres et al. (2019), which estimates changes in trade costs associated with the BRI. Although differences in methodology complicate a direct comparison, the reductions we estimate for IMEC are smaller in magnitude, with far less global impact than the BRI. These differences are consistent with the much broader scope of the BRI, which consists of over 50 infrastructure projects spanning a considerably larger geographic area. When attention is restricted to directly affected countries, however, the magnitudes of the estimated trade-cost reductions become more comparable. In this sense, much of the difference between the estimated effects of IMEC and the BRI reflects differences in their geographic reach rather than fundamentally different effects among the countries most directly exposed.

4.2 Trade and Welfare

We now proceed to the GE trade and welfare implications of IMEC. In contrast to the PE trade cost analysis, these results incorporate endogenous adjustments in each country's containerized output, expenditures, prices, and multilateral resistance terms, as defined by Equations (1)–(5). Thus, changes in bilateral trade are influenced, but not entirely determined, by the patterns discussed in the previous section. Most notably, while IMEC does not raise any bilateral trade costs in levels, it can make shipping between a given country pair relatively more expensive. For example, a country whose bilateral trade costs remain unchanged may become a relatively less competitive supplier to a destination if IMEC substantially lowers that destination's trade costs with other exporters. These changes in relative trade costs are captured by the multilateral resistance terms and generate both trade creation and trade diversion in general equilibrium.

First, we assess the extent to which IMEC facilitates trade among its participants. Table 6 reports



Figure 2: Simple average decrease in containerized trade costs by exporter in the central scenario. Darker shades denote greater reductions in trade costs.

percentage changes in containerized and aggregate trade for IMEC participants in our central scenario, abstracting from intranational production. We obtain changes in aggregate trade flows by combining our results for containerized goods with all other sectors in the ITPD-S that lie outside IMEC's scope.[16] Trade growth is strongest within the Arabian Peninsula, where IMEC enhances the region's overland connectivity. Intra-GCC trade growth is particularly strong, as the corridor improves the relative accessibility of GCC markets to one another. The corridor also substantially expands trade between the GCC countries and Israel and Jordan. Beyond the region, GCC exports to the European Union and United States increase by approximately 9% and 12%, respectively.

The trade responses for India are mixed. Containerized exports to the United States grow by approximately 19%, which translates to an aggregate increase of 7.5%. Export growth to the GCC is heavily driven by Saudi Arabia, the only GCC country for which bilateral trade costs with India decline. Exports to the other five GCC countries fall by approximately 12% on average, as India becomes relatively less competitive in these markets. As discussed above, this asymmetry reflects the geography of the region: Saudi Arabia is the only GCC country with major port infrastructure on the Red Sea, allowing the rail segment of IMEC to shorten the route between India and its western coast. By contrast, the corridor provides little advantage for trade with the other GCC countries. India's imports from the European Union also increase modestly, whereas its exports to the European Union decline. These results reinforce the partial-equilibrium finding that IMEC does not generally provide a sufficiently large cost advantage to substantially redirect India–Europe trade away from existing maritime routes.

[16]Notably, all additional sectors are held fixed. This exercise is meant to contextualize the results for containerizable goods. We acknowledge that a proper analysis of the implications for aggregate trade requires a multisector approach.

Table 6: IMEC's Impact on Bilateral Trade for Selected Countries

Exporter / Importer	EU	GCC	IND	ISR	JOR	USA
Containerized Trade						
EU	−0.1	16.7	1.0	0.0	19.8	−0.7
GCC	23.1	137.1	−0.5	18.4	12.3	29.8
IND	−2.0	30.8		−1.7	−14.0	19.4
ISR	−1.8	53.3	−0.4		−0.1	−2.0
JOR	45.0	46.4	−22.8	8.6		−19.5
USA	0.3	20.2	13.2	0.6	−7.3	
Aggregate Trade						
EU	−0.1	12.3	0.6	0.0	14.8	−0.3
GCC	9.2	44.8	0.0	18.4	2.0	11.5
IND	−1.2	15.4		−1.2	−6.5	7.5
ISR	−1.0	53.3	−0.1		−0.1	−0.4
JOR	41.8	21.5	−5.0	6.0		−10.9
USA	0.1	6.7	2.5	0.1	−2.0	

Note: Table reports percentage changes in bilateral trade flows. We construct aggregate trade changes by merging the results for containerized goods with all other sectors in the ITPD-S, where the size of all non-containerized sectors is held fixed.

Figure 3 displays changes in containerized exports at the country level. Export decreases are displayed in light red, negligible changes (those with an absolute value smaller than a tenth of a percent) are shown in white, and export growth is shown in varying shades of blue. In total, world containerized trade rises by 0.5%. Among countries experiencing export declines, the effects are modest, averaging -0.11% with a median of -0.07%. The Marshall Islands experience the largest decline in containerized exports at -0.4%. As with trade costs, export growth is concentrated in the GCC countries, with small gains accruing to countries in Europe, North America, and North Africa. Containerized export growth is strongest in the UAE at approximately 49%, followed by Saudi Arabia and Kuwait at around 16%, and Qatar and Bahrain at around 12%. Cyprus and Jordan also experience export growth of roughly 10%. Despite experiencing limited bilateral trade-cost reductions with other IMEC participants, India's containerized exports increase by 1.7%.

Figure 4 similarly depicts changes in containerized real GDP, our measure of welfare in the counterfactual simulation. Again, decreases are shown in red, negligible changes in white, and increases in varying shades of blue. Global containerized real GDP rises by 0.03%. Among the countries in our sample, 127 experience declines in containerized real GDP, with an average decline of 0.06%. The largest decreases are concentrated among countries along the Red Sea that are bypassed by the IMEC route, including Yemen (-0.23%), Eritrea (-0.17%), Djibouti (-0.15%), and Sudan (-0.14%). Growth in containerized real GDP is strongest in Cyprus (5.27%), followed by Jordan (4.9%). Each of the GCC countries experiences growth in the range of 2–4%, except for Oman (1.1%). Other economies associated with the corridor experience

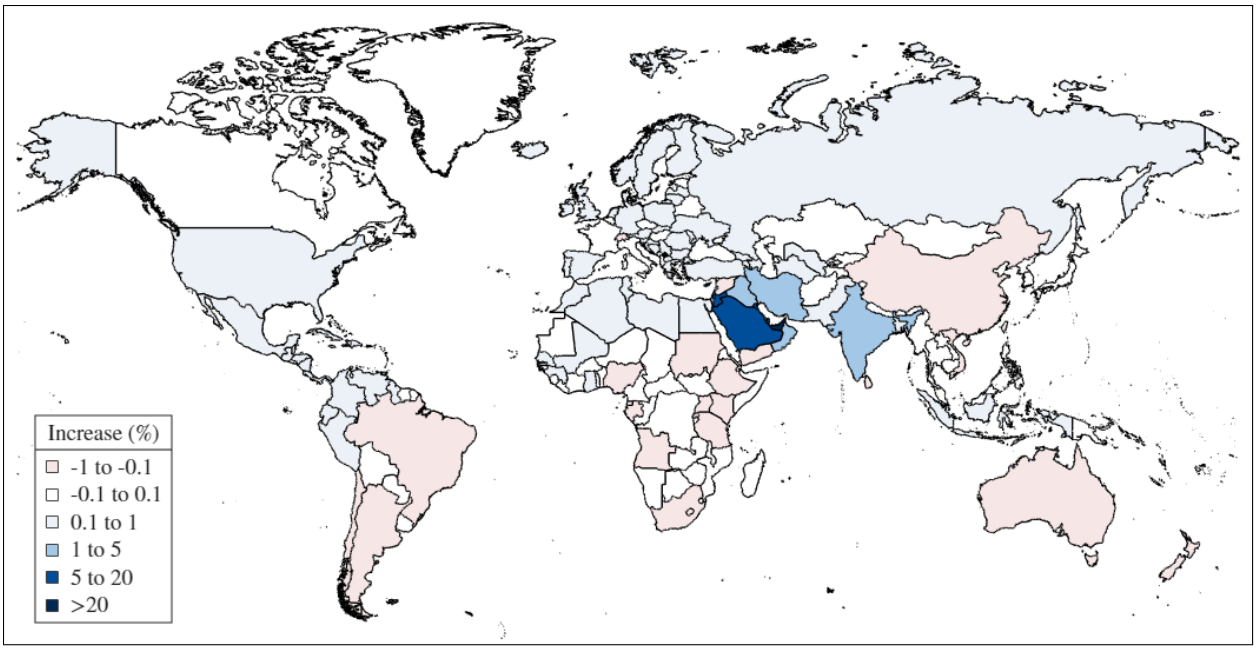


Figure 3: Change in containerized exports by country.

comparatively small gains, with the EU, India, Israel, and the United States each recording increases of less than 0.2%.

Last, given the GCC countries' focus on reducing reliance on hydrocarbons (as stated in, e.g., Saudi Vision 2030 and Abu Dhabi Economic Vision 2030), we assess the extent to which IMEC alters the composition of GCC exports in our central scenario. Figure 5 displays aggregate hydrocarbon and manufacturing export shares for the GCC countries. The baseline values reflect sector shares in the ITPD-S, while the IMEC values reflect our counterfactual exercise. The GCC countries experience modest export diversification as a result of the new trade route, with the share of hydrocarbon exports falling by 3.6 percentage points and the share of manufacturing exports rising by a similar amount. This shift is primarily driven by the UAE, whose share of hydrocarbon exports falls by approximately 6 percentage points. Other GCC countries experience smaller declines, with hydrocarbon shares falling by 1 to 2 percentage points in Saudi Arabia, Kuwait, and Qatar, and by negligible amounts in Oman and Bahrain. These results suggest that IMEC may contribute incrementally to ongoing diversification efforts but is unlikely to substantially alter the composition of GCC exports.

4.3 Sensitivity and Corridor Viability

Our upper- and lower-bound scenarios again emphasize the importance of relative rail costs and the value of time savings in determining the effectiveness of IMEC. Most notably, these assumptions affect not only the magnitude of the corridor's gains but also the set of bilateral relationships for which IMEC is economically viable. As shown in Table 7, the number of bilateral pairs for which IMEC offers a lower trade cost than the Suez Canal rises from 374 in the lower-bound scenario, to 1,974 in the central scenario, and to 3,580

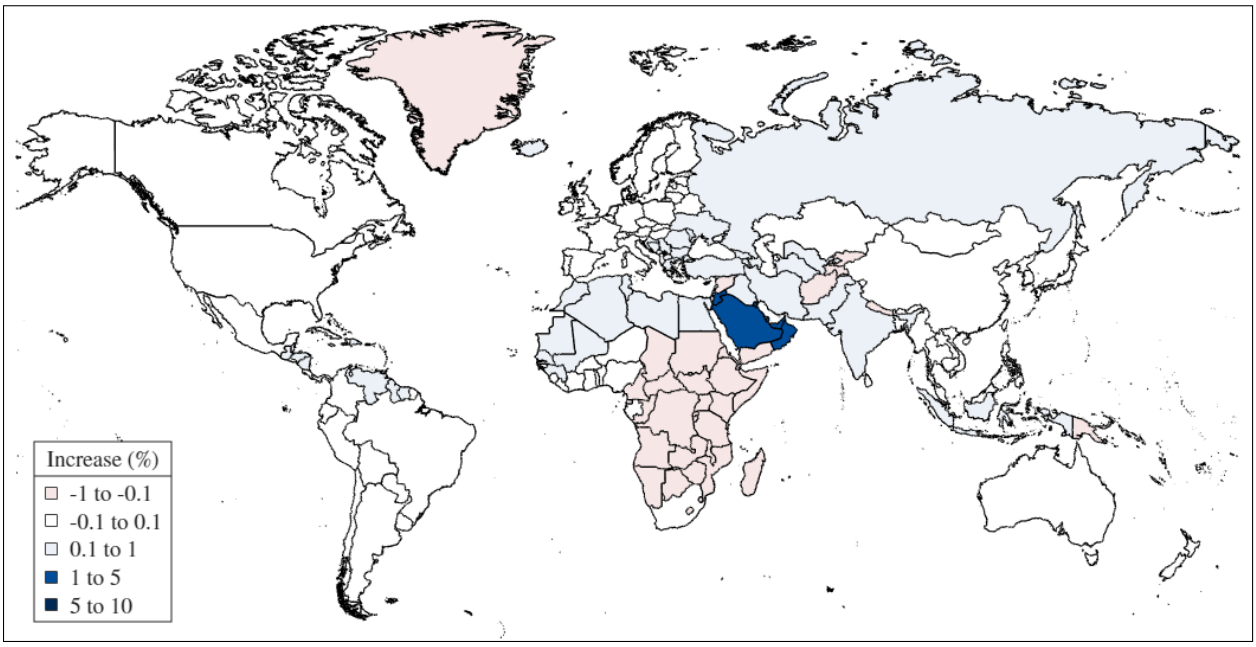


Figure 4: Change in containerized real GDP by country.

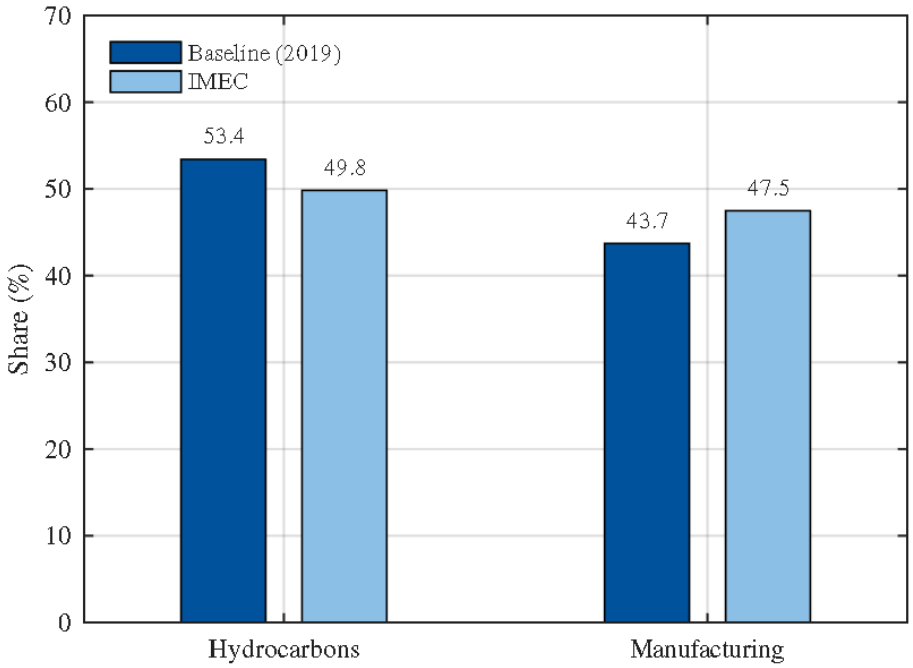


Figure 5: GCC export diversification for selected sectors. Note that hydrocarbon exports are held fixed in levels, but vary as a share of total exports.

in the upper-bound scenario. This nonlinear response reflects our assumption that IMEC and the Suez Canal are perfect substitutes: once the cost of using IMEC falls below that of the Suez route, shipments switch discretely to the corridor. As a result, changes in the relative attractiveness of IMEC operate not only on the intensive margin, by changing the size of trade-cost reductions, but also on the extensive margin, by determining which bilateral routes adopt IMEC at all. These differences in route adoption translate into substantial differences in the gains accruing to countries located along the corridor. Across the lower-bound, central, and upper-bound scenarios, exports from the IMEC core increase by approximately 6.8%, 22.1%, and 36.9%, respectively, while containerized real GDP rises by 0.8%, 2.4%, and 4.4%. The corresponding global effects are considerably smaller: world containerized trade rises by approximately 0.09%, 0.51%, and 0.91%, while global containerized real GDP increases by 0.01%, 0.03%, and 0.06%. Thus, more favorable rail costs and time savings substantially broaden the reach and regional benefits of IMEC, even though its effects on the global economy remain comparatively modest.

The viability of IMEC is similarly sensitive to delays along the corridor. Table 7 displays the effects of introducing delays of varying length into our counterfactual exercise. One-, two-, and three-day delays reduce the number of bilateral pairs for which IMEC offers a cost advantage over the Suez route from 1,974 in the central scenario to 1,905, 1,753, and 1,568, respectively. The effects become more pronounced for longer delays. With a seven-day delay, only 137 bilateral pairs across 36 countries find IMEC less costly than the Suez route. A ten-day delay further narrows participants to 73 bilateral pairs across 23 countries. The declining route adoption also confers fewer benefits to countries along the corridor. IMEC-core export growth falls from 22.1% in the central scenario to 9.0% with a seven-day delay and 7.8% with a ten-day delay. The corresponding containerized real GDP gains fall from 2.4% to approximately 1.0% and 0.8%. At the global level, the increase in containerized trade falls from 0.51% in the central scenario to approximately 0.11% and 0.09% under the seven- and ten-day delays, respectively. Taken together, these results emphasize that the economic viability of IMEC depends on two dimensions: the corridor must offer both competitive freight rates and transit times to maintain a logistical advantage over existing maritime routes.

Given the sensitivity of our results to these assumptions, we also estimate the degree of cost efficiency IMEC would need to achieve to generate economically meaningful gains. As a first illustrative benchmark, we calculate the rail-to-sea freight ratio required to raise aggregate containerized real GDP among the IMEC-core countries by 5%. These countries are likely to bear a substantial share of the corridor's infrastructure requirements because the route directly traverses several of their territories. As a second benchmark, we ask what freight ratio would generate approximately $30 billion in additional annual containerized real GDP, equivalent to roughly 8% of baseline containerized GDP in the IMEC core.[17] With a value-of-time parameter of 2.1%, we estimate that rail freight rates can be at most 47% higher than maritime rates for IMEC to raise containerized real GDP in the core countries by 5%. Reaching the larger $30 billion benchmark

[17]Many news reports place total IMEC costs around $20 billion, though this figure is unsubstantiated. Based on cost assumptions from De Soyres, Mulabdic and Ruta (2020), $30 billion corresponds approximately to the cost of constructing 2,500 kilometers of new rail. Although not all railways would need to be built from scratch, additional expenses may arise from harmonizing existing lines across borders and upgrading port infrastructure. Because this construction-cost estimate is a one-time capital cost whereas our real-GDP gains are annual flows, the comparison should be interpreted only as an illustrative benchmark rather than a formal cost-benefit analysis. We strongly encourage a more rigorous cost assessment for policy discussions.

Table 7: Sensitivity of IMEC Outcomes to Freight Rates and Transit Delays

Scenario	Bilateral Pairs	Countries	IMEC Core Exports (%)	IMEC Core RGDP (%)	World Exports (%)	World RGDP (%)
Lower	374	121	6.82	0.80	0.09	0.01
Preferred	1,974	150	22.14	2.45	0.51	0.03
Upper	3,580	159	36.85	4.44	0.91	0.06
1 Day Delay	1,905	141	19.38	2.11	0.42	0.03
2 Day Delay	1,753	139	16.79	1.81	0.33	0.02
3 Day Delay	1,568	129	14.48	1.55	0.25	0.02
7 Day Delay	137	36	9.02	0.99	0.11	0.01
10 Day Delay	73	23	7.75	0.83	0.09	0.01

Note: "IMEC Core" includes Israel, Jordan, Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates. "Bilateral Pairs" counts bilateral routes for which the IMEC route has a lower route-specific trade cost than the Suez Canal. "Countries" counts unique countries appearing in at least one bilateral pair for which IMEC lowers aggregate bilateral trade costs, which additionally requires positive baseline Suez exposure. Export and real GDP (RGDP) outcomes are percentage changes relative to the baseline. IMEC-core outcomes aggregate across core economies before calculating percentage changes.

requires rail rates to be no more than approximately 20% above sea freight rates. Under the lower value-of-time parameter of 0.6%, the corresponding rail-to-sea freight ratios tighten to approximately 1.36 and 1.08, respectively. These thresholds illustrate how strongly the corridor's economic gains depend on maintaining competitive rail costs, particularly when the value placed on time savings is relatively low. Our analysis also abstracts from fees that may be levied along the route, the revenues from which could help finance infrastructure investment but would also raise the cost of using the corridor.

4.4 Operational considerations and limitations

This section interprets our results in light of the assumptions underlying our analytical framework. First, our analysis assumes that rail freight can move between Haifa and Dubai without substantial delays when crossing international borders. While customs inspections are included in port processing times in Israel and the UAE, we abstract from time costs associated with border compliance at the Israel–Jordan, Jordan–Saudi Arabia, and Saudi Arabia–UAE borders. This assumption provides a useful benchmark for the corridor, but depends critically on multilateral coordination. Using border-compliance times from the World Bank's Doing Business Database suggests that these frictions could add as much as three days to transit along the IMEC route, substantially eroding its time advantage. Consistent with the sensitivity analysis in Table 7, delays of this magnitude materially reduce both the number of bilateral routes for which IMEC is competitive with the Suez Canal and the resulting trade and welfare gains.

Our counterfactual also abstracts from several other operational and institutional frictions that could affect corridor usage. These include geopolitical risk and associated insurance costs, differences in legal

and regulatory frameworks, the allocation of transit liability, customs interoperability and associated data-sharing requirements, transshipment and drayage costs associated with transferring cargo between ship, truck, and rail, and heterogeneity in the value placed on delivery time across different types of cargo. Incorporating these features could either raise the generalized cost of using IMEC or alter the set of shipments for which the corridor is competitive with existing maritime routes.

Second, for tractability, we assume that IMEC and the Suez Canal are perfect substitutes for containerized shipments through the Middle East, with deterministic route choice based on generalized trade costs. In practice, substitution between the two routes is likely to depend on additional factors beyond the scope of our model. Each route, for example, is exposed to distinct risks. In recent years, the Suez Canal has experienced blockages, while disruptions in the Red Sea have substantially reduced traffic through the route. IMEC would also not be immune to disruptions. In particular, conflict escalation could disrupt corridor operations, including through infrastructure damage, while broader geopolitical risks could affect the reliability of the route. The two routes may also differ in their exposure to congestion and capacity constraints. If rail capacity or port throughput emerges as a binding constraint, IMEC may be less attractive than the Suez Canal even when it offers lower point-to-point trade costs.

Finally, our analysis quantifies the potential trade and welfare gains from IMEC but does not provide a comprehensive cost-benefit assessment. Constructing, operating, and maintaining the corridor would require substantial fixed and variable expenditures on rail, port, and supporting infrastructure, and these costs may be distributed unevenly across participating countries. Positive trade and welfare effects are therefore not sufficient to establish that IMEC is economically justified. The relevant policy comparison must also account for the opportunity cost of alternative investments in transportation infrastructure, trade facilitation, or other development priorities, as well as implications for fiscal and debt sustainability. Moreover, the distribution of infrastructure costs and financing responsibilities across participating countries remains uncertain. Without credible cost estimates and a clear allocation of financial responsibility, it is difficult to determine whether IMEC represents an efficient use of public resources.

5 Conclusion

We provide the first quantitative assessment of the IMEC trade route. Three key findings emerge from our counterfactual. First, IMEC would generate its largest gains for countries located on or near the Arabian Peninsula, particularly the GCC economies, which would benefit from improved overland connectivity and reduced distances to countries in Europe, North America, and North Africa. Second, trade costs involving countries in South America, Sub-Saharan Africa, and East Asia would be largely unaffected by the agreement, leading to some trade diversion as these regions become relatively more costly trading partners for countries in the Middle East. Third, the corridor's effectiveness hinges crucially on the relative price of maritime and rail freight. This tradeoff is especially important for trade between India and the European Union, where time and distance savings are often insufficient to offset the higher cost of rail transport.

At the same time, the results should be interpreted with caution. Limited clarity regarding implementation, financing, freight capacity, and regulatory integration complicates a complete assessment of IMEC.

Our findings suggest that the benefits generated by the corridor could be economically meaningful relative to the investment required, but we strongly emphasize that a rigorous cost-benefit analysis is essential before drawing firm policy conclusions. Such an analysis would require more concrete information on the proposed route, construction and maintenance costs, financing, and implementation timeline. These details would allow for a better assessment of whether the trade and welfare benefits of the corridor are sufficient to justify the scale of investment required.

Moreover, our estimates do not capture the entire economic value that IMEC could provide. We explicitly model IMEC as competing with existing maritime routes in the absence of disruptions. Any benefits the corridor provides when those routes are impaired or unavailable therefore lie outside the scope of our paper. Quantifying these additional benefits would require explicit assumptions about the probability, duration, and severity of future disruptions. Furthermore, we focus on the containerizable goods sector and hold all other sectors fixed. Our results therefore abstract from cross-sector income and substitution effects that could alter the aggregate gains from IMEC. Future research could incorporate these extensions to provide a more comprehensive assessment of corridor viability.

References

Abu-Aisha, Tareq, Jean-François Audy, and Mustapha Ouhimmou. 2024. "Toward an efficient sea-rail intermodal transportation system: a systematic literature review." *Journal of Shipping and Trade*, 9(1): 23.

Allen, Treb, and Costas Arkolakis. 2022. "The Welfare Effects of Transportation Infrastructure Improvements." *The Review of Economic Studies*, 89(6): 2911–2957.

Anderson, James E, and Eric Van Wincoop. 2003. "Gravity with Gravitas: A Solution to the Border Puzzle." *American Economic Review*, 93(1): 170–192.

Anderson, James E., and Eric van Wincoop. 2004. "Trade Costs." *Journal of Economic Literature*, 42(3): 691–751.

Baccelli, Oliviero, and Pietro Morino. 2020. "The role of port authorities in the promotion of logistics integration between ports and the railway system: The Italian experience." *Research in Transportation Business & Management*, 35: 100451.

Baier, Scott L., and Jeffrey H. Bergstrand. 2007. "Do free trade agreements actually increase members' international trade?" *Journal of International Economics*, 71(1): 72–95.

Bernhofen, Daniel M., Zouheir El-Sahli, and Richard Kneller. 2016. "Estimating the effects of the container revolution on world trade." *Journal of International Economics*, 98: 36–50.

Bertoli, Simone, Michaël Goujon, and Olivier Santoni. 2016. "The CERDI-seadistance database." *Etudes et Documents*.

Bird, Julia, Mathilde Lebrand, and Anthony J. Venables. 2020. "The Belt and Road Initiative: Reshaping economic geography in Central Asia?" *Journal of Development Economics*, 144: 102441.

Borchert, Ingo, Mario Larch, Serge Shikher, and Yoto V. Yotov. 2021. "The International Trade and Production Database for Estimation (ITPD-E)." *International Economics*, 166: 140–166.

Borchert, Ingo, Mario Larch, Serge Shikher, and Yoto V. Yotov. 2024. "Globalization, Trade, and Inequality: Evidence from a New Database." United States International Trade Commission USITC Working Paper 2024–11–A.

Broda, Christian, Joshua Greenfield, and David Weinstein. 2006. "From Groundnuts to Globalization: A Structural Estimate of Trade and Growth."

Caliendo, Lorenzo, and Fernando Parro. 2015. "Estimates of the Trade and Welfare Effects of NAFTA." *The Review of Economic Studies*, 82(1): 1–44.

Chao, Shih-Liang, Ming-Miin Yu, and Wei-Fan Hsieh. 2018. "Evaluating the efficiency of major container shipping companies: A framework of dynamic network DEA with shared inputs." *Transportation Research Part A: Policy and Practice*, 117: 44–57.

Chaudhury, Dipanjan Roy. 2024. "IMEC: Indian Team in UAE Discusses Start of Work on Trade Route." *The Economic Times*.

Chu, Chi-Hsiang, and Nina Ledermann. 2017. "Panama Canal Utilization." Center for Transportation Research, University of Texas at Austin Transportation policy brief #1.

DD India. 2026. "India-UAE Ties Deepen as Trade Crosses $100 Billion: Indian Envoy to UAE Deepak Mittal Ahead of PM Modi's Visit." *DD India*.

De Soyres, François, Alen Mulabdic, and Michele Ruta. 2020. "Common transport infrastructure: A quantitative model and estimates from the Belt and Road Initiative." *Journal of Development Economics*, 143: 102415.

De Soyres, François, Alen Mulabdic, Siobhan Murray, Nadia Rocha, and Michele Ruta. 2019. "How much will the Belt and Road Initiative reduce trade costs?" *International Economics*, 159: 151–164.

Donaldson, Dave. 2018. "Railroads of the Raj: Estimating the Impact of Transportation Infrastructure." *American Economic Review*, 108: 899–934.

Donaldson, Dave, and Richard Hornbeck. 2016. "Railroads and American Economic Growth: A "Market Access" Approach *." *The Quarterly Journal of Economics*, 131(2): 799–858.

Eaton, Jonathan, and Samuel Kortum. 2002. "Technology, Geography, and Trade." *Econometrica*, 70(5): 1741–1779.

Egger, Peter, and Mario Larch. 2008. "Interdependent preferential trade agreement memberships: An empirical analysis." *Journal of International Economics*, 76(2): 384–399.

Egger, Peter, Mario Larch, Sergey Nigai, and Yoto Yotov. 2021. "Trade costs in the global economy: Measurement, aggregation and decomposition." WTO Staff Working Paper Working Paper ERSD-2021-2.

Emirates News Agency. 2026. "Gulftainer Unveils US$2 Billion Global Trade Infrastructure Strategy."

European Commission. 2023. "Speech 23 4426: State of the Union address." *https://ec.europa.eu/commission/presscorner/detail/en/speech_23_4426*, European Commission Press Corner.

Feyrer, James. 2019. "Trade and Income—Exploiting Time Series in Geography." *American Economic Journal: Applied Economics*, 11(4): 1–35.

Feyrer, James. 2021. "Distance, trade, and income — The 1967 to 1975 closing of the Suez canal as a natural experiment." *Journal of Development Economics*, 153: 102708.

Fontagné, Lionel, Houssein Guimbard, and Gianluca Orefice. 2022. "Tariff-based product-level trade elasticities." *Journal of International Economics*, 137: 103593.

Fredman, Asher, and Joseph Rozen. 2025. "The India - Middle East - Europe Economic Corridor." *Misgrav Institute for National Security*.

Ganapati, Sharat, Woan Foong Wong, and Oren Ziv. 2024. "Entrepôt: Hubs, Scale, and Trade Costs." *American Economic Journal: Macroeconomics*, 16(4): 239–278.

Guimbard, Houssein, Sébastien Jean, Mondher Mimouni, and Xavier Pichot. 2012. "MAcMap-HS6 2007, An exhaustive and consistent measure of applied protection in 2007." *International Economics*, 130: 99–121.

Gulf Cooperation Council General Secretariat. 2026. "GCC Ministers of Transport and Communications Discuss Current Situation and Its Impact on Supply Chains, the Readiness of Land Border Crossings, and Ways to Address Operational Challenges."

Gurevich, Tamara, and Peter Herman. 2018. "The Dynamic Gravity Dataset: 1948–2016." United States International Trade Commission USITC Working Paper 2018–02–A.

Hoffmeister, Onno, Ronald Halim, Nicholas-Joseph Lazarou, Nour Barnat, David Cristallo, Dominik Englert, and Jan Hoffmann. 2022. "Developing a global transport costs dataset for international trade." United Nations Conference on Trade and Development UNCTAD Research Paper No. 85.

Hummels, David. 2007. "Transportation Costs and International Trade in the Second Era of Globalization." *Journal of Economic Perspectives*, 21(3): 131–154.

Hummels, David L, and Georg Schaur. 2013. "Time as a Trade Barrier." *American Economic Review*, 103(7): 2935–2959.

John A. Volpe National Transportation Systems Center. 2021. "National Transportation Statistics 2021: 50th Anniversary Edition." U.S. Department of Transportation, Bureau of Transportation Statistics.

Larch, Mario, Serge Shikher, and Yoto V. Yotov. 2025. "The International Trade and Production Database for Estimation—Release 3 (ITPD-E-R03)." United States International Trade Commission USITC Working Paper 2025–06–A.

Press Information Bureau Delhi. 2025. "India poised to become a trusted bridge of global connectivity through India–Middle East–Europe Economic Corridor (IMEC): Shri Piyush Goyal." *https://www.pib.gov.in/PressReleasePage.aspx?PRID=2122299&reg=3&lang=2*, Press Information Bureau, Government of India.

Rose, Andrew K. 2004. "Do We Really Know That the WTO Increases Trade?" *American Economic Review*, 94: 98–114.

Slack, Brian, Claude Comtois, Bart Wiegmans, and Patrick Witte. 2017. "Ships time in port." *International Journal of Shipping and Transport Logistics*.

Suri, Navdeep, Nilanjan Ghosh, Kabir Taneja, Sameer Patil, and Promit Mookherjee. 2024. "India-Middle East-Europe Economic Corridor." *Observer Research Foundation*.

United Nations Conference on Trade and Development. 2025*a*. *Review of Maritime Transport 2025.* . 2025 ed., United Nations.

United Nations Conference on Trade and Development. 2025*b*. *The Trade-and-Transport Dataset: Technical Documentation.* Geneva:United Nations. OCLC: 1558295722.

Verschuur, Jasper, Johannes Lumma, and Jim W. Hall. 2025. "Systemic impacts of disruptions at maritime chokepoints." *Nature Communications*, 16(1): 10421.

Verschuur, Jasper, Seung Mo Choi, Parisa Kamali, Robin Koepke, Matthew McKetty, Michele Ruta, Mario Saraiva, Alessandra Sozzi, and Serkan Arslanalp. 2025. "Nowcasting Global Trade from Space."

Yotov, Yoto V., Roberta Piermartini, José-Antonio Monteiro, and Mario Larch. 2016. *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model.* WTO.

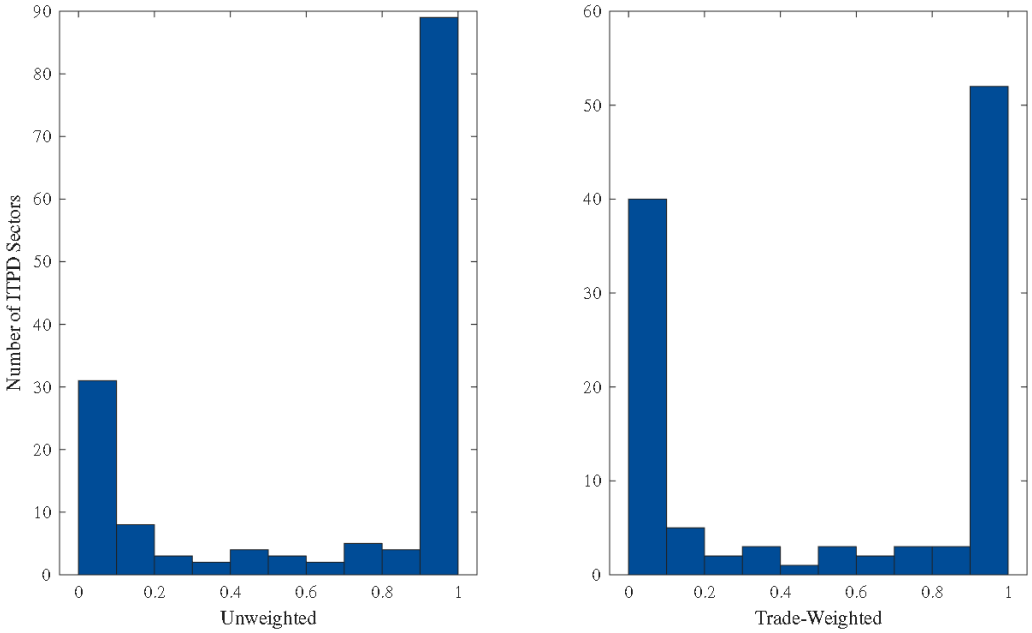


Figure A.1: Average containerizability by ITPD sector.

A Containerizable Goods

In this section, we describe our process for labeling sectors in the ITPD-E and ITPD-S as containerizable. We begin with the list provided by Bernhofen, El-Sahli and Kneller (2016), which is based on classifications from the Verband deutscher Ingenieure (German Association of Engineers). Goods are categorized into three groups by 4-digit Standard International Trade Classification (SITC) product codes: fully containerizable, partially containerizable, and noncontainerizable. To maintain conservative estimates for IMEC, we restrict our attention to those goods that are most containerizable, which are displayed in Table A.1.

Because no direct correspondence exists between the SITC and ITPD nomenclature, we first map four-digit SITC codes to six-digit Harmonized System (HS) codes and FAOSTAT Commodity List (FCL) codes for agricultural goods. Using the International Standard Industrial Classification (ISIC) nomenclature, we then map HS codes to ITPD sectors. All goods without a clear mapping from SITC codes to ITPD sectors are marked as noncontainerizable. To aggregate product-level containerizability to the sector level, we calculate simple and trade-weighted averages for each sector. Trade weights are derived from CEPII BACI product-level trade flows across all country pairs in 2019.

Figure A.1 displays histograms of the sector-level containerizability. For both simple and weighted measures, the majority of ITPD sectors are above 90% containerizable or below 10%, which leaves few ambiguous cases. Conservatively, we label sectors as containerizable if greater than 80% of the underlying products are containerizable in both the weighted and unweighted averages, though this threshold has little consequence for our results. Table A.2 presents our final list of containerizable goods sectors.

Table A.1: Class A Containerizable Products (4-digit SITC, 1968)

SITC Code	Product description	No. of products
035	Fish, dried, salted or in brine; smoked fish	1
037	Fish, crustaceans, and molluscs, prepared or preserved	3
042	Rice	3
046	Meal and flour of wheat and flour of meslin	1
047	Other cereal meals and flours	1
048	Cereal preparations and preparations of flour of fruits or vegetables	6
056	Vegetables, roots, and tubers, prepared or preserved, n.e.s.	4
058	Fruit, preserved, and fruit preparations	1
061	Sugar and honey	6
062	Sugar confectionery and other sugar preparations	1
071	Coffee and coffee substitutes	3
072	Cocoa	4
073	Chocolate and other food preparations containing cocoa	1
074	Tea and mate	3
075	Spices	3
081	Feeding stuff for animals (excluding unmilled cereals)	6
091	Margarine and shortening	3
098	Edible products and preparations, n.e.s.	1
111	Non-alcoholic beverages, n.e.s.	1
112	Alcoholic beverages	5
121	Tobacco, unmanufactured; tobacco refuse	4
122	Tobacco, manufactured	4
211	Hides and skins (except furskins), raw	7
212	Furskins, raw (including astrakhan, caracul, etc.)	1
222	Oil seeds and oleaginous fruits (excluding flours and meals)	7
223	Oil seeds and oleaginous fruits, whole or broken (incl. flours and meals)	7
23	Crude rubber (including synthetic and reclaimed)	5
244	Cork, natural, raw and waste (incl. blocks and sheets)	1
25	Pulp and waste paper	7
26	Textile fibres (except wool tops) and their wastes	30
277	Natural abrasives, n.e.s. (including industrial diamonds)	3
291	Crude animal materials, n.e.s.	3

Continued on next page

SITC Code	Product description	No. of products
411	Animal oils and fats	3
423	Fixed vegetable oils, soft, crude or refined	7
424	Other fixed vegetable oils, fluid or solid, crude	7
431	Animal and vegetable oils and fats, processed; waxes	5
53	Dyeing, tanning, and colouring materials	11
54	Medicinal and pharmaceutical products	8
55	Essential oils, perfumes, toilet and cleansing preparations	9
58	Artificial resins and plastic materials; cellulose esters and ethers	29
59	Chemical materials and products, n.e.s.	13
61	Leather, leather manufactures, n.e.s.; dressed furskins	13
62	Rubber manufactures, n.e.s.	13
63	Cork and wood manufactures (excluding furniture)	13
64	Paper, paperboard, and articles of paper or pulp	15
65	Textile yarn, fabrics, made-up articles, and related products	61
664	Glass	10
665	Glassware	5
666	Pottery	1
667	Pearls, precious and semi-precious stones	5
673	Iron and steel bars, rods, angles, and sections	5
674	Universals, plates, and sheets of iron or steel	8
675	Hoop and strip of iron or steel	1
677	Iron or steel wire, coated or not, not insulated	1
678	Tubes, pipes, and fittings of iron or steel	6
679	Iron and steel castings, forgings, and stampings, rough	1
681	Silver, platinum, and other platinum-group metals	3
682	Copper	3
683	Nickel	3
684	Aluminium	3
685	Lead	3
686	Zinc	3
687	Tin	3
689	Miscellaneous non-ferrous base metals for metallurgy	3
692	Metal containers for storage or transport	3
693	Wire products and fencing grills	4
694	Nails, screws, nuts, bolts, and similar articles	1
695	Tools for use in hand or in machines	5

Continued on next page

SITC Code	Product description	No. of products
696	Cutlery	2
697	Household equipment of base metal, n.e.s.	5
699	Manufactures of base metal, n.e.s.	10
71	Power-generating machinery and equipment	26
723	Civil engineering and contractors' plant and parts	4
724	Textile and leather machinery and parts	7
725	Paper and pulp mill machinery	4
726	Printing and bookbinding machinery and parts	6
727	Food-processing machinery and parts	2
728	Machinery and equipment specialized for particular industries	5
73	Metalworking machinery	11
741	Heating and cooling equipment and parts	6
742	Pumps for liquids and liquid elevators	7
743	Pumps, compressors, fans, blowers, centrifuges	8
7449	Parts of machinery of heading 744.2	1
745	Other non-electrical machine tools and parts	2
749	Non-electric parts and accessories of machines	5
75	Office machines and automatic data-processing equipment	15
76	Telecommunications and sound recording apparatus	16
77	Electrical machinery and apparatus, n.e.s.	31
7840	Parts and accessories of motor vehicles (headings 722–783)	1
7929	Parts of aircraft, excluding tires and engines	1
8	Miscellaneous manufactured articles	114

Note: Panel A lists Class A containerizable products at the 4-digit SITC (Rev. 1, 1968) level. Numbers indicate the count of underlying 4-digit product categories within each heading.

Table A.2: Containerizable ITPD Sectors

ITPD Sector	Product description
2	Rice (raw)
5	Cereal products
10	Other sweeteners
15	Prepared vegetables

Continued on next page

ITPD Sector	Product description
21	Cocoa and cocoa products
22	Beverages, nec
23	Cotton
24	Tobacco leaves and cigarettes
25	Spices
41	Grain mill products
42	Starches and starch products
43	Prepared animal feeds
44	Bakery products
45	Sugar
46	Cocoa chocolate and sugar confectionery
47	Macaroni noodles & similar products
50	Wines
51	Malt liquors and malt
52	Soft drinks; mineral waters
53	Tobacco products
54	Textile fibre preparation; textile weaving
55	Made-up textile articles except apparel
56	Carpets and rugs
57	Cordage rope twine and netting
58	Other textiles n.e.c.
59	Knitted and crocheted fabrics and articles
60	Wearing apparel except fur apparel
61	Dressing & dyeing of fur; processing of fur
63	Luggage handbags etc.; saddlery & harness
64	Footwear
66	Veneer sheets plywood particle board etc.
67	Builders' carpentry and joinery
68	Wooden containers
69	Other wood products; articles of cork/straw
70	Pulp paper and paperboard
71	Corrugated paper and paperboard
72	Other articles of paper and paperboard
73	Publishing of books and other publications
74	Publishing of newspapers journals etc.
76	Other publishing
77	Printing

Continued on next page

ITPD Sector	Product description
78	Service activities related to printing
84	Plastics in primary forms; synthetic rubber
86	Paints varnishes printing ink and mastics
87	Pharmaceuticals medicinal chemicals etc.
88	Soap cleaning & cosmetic preparations
89	Other chemical products n.e.c.
90	Man-made fibres
91	Rubber tyres and tubes
92	Other rubber products
93	Plastic products
94	Glass and glass products
103	Basic precious and non-ferrous metals
105	Tanks reservoirs and containers of metal
106	Steam generators
107	Cutlery hand tools and general hardware
109	Engines & turbines (not for transport equipment)
110	Pumps compressors taps and valves
111	Bearings gears gearing & driving elements
113	Lifting and handling equipment
114	Other general purpose machinery
115	Agricultural and forestry machinery
116	Machine tools
117	Machinery for metallurgy
118	Machinery for mining & construction
119	Food/beverage/tobacco processing machinery
120	Machinery for textile apparel and leather
122	Other special purpose machinery
123	Domestic appliances n.e.c.
124	Office accounting and computing machinery
127	Insulated wire and cable
129	Lighting equipment and electric lamps
133	TV and radio receivers and associated goods
134	Medical surgical and orthopaedic equipment
136	Optical instruments & photographic equipment
137	Watches and clocks
138	Motor vehicles
139	Automobile bodies trailers & semi-trailers

Continued on next page

ITPD Sector	Product description
140	Parts/accessories for automobiles
141	Building and repairing of ships
142	Building/repairing of pleasure/sport. boats
143	Railway/tramway locomotives & rolling stock
144	Aircraft and spacecraft
145	Motorcycles
146	Bicycles and invalid carriages
147	Other transport equipment n.e.c.
148	Furniture
149	Jewellery and related articles
150	Musical instruments
151	Sports goods
152	Games and toys
153	Other manufacturing n.e.c.

B Additional Results

Table A.3: Changes in Trade Costs Associated with the Belt and Road Initiative

Group	Min (%)	Max (%)	Mean (%)	Std Dev (%)
World – Lower Bound	0.00	61.52	1.05	2.43
World – Upper Bound	0.00	65.16	2.19	3.40
BRI Countries – Lower Bound	0.00	61.52	1.50	3.07
BRI Countries – Upper Bound	0.00	65.16	2.81	4.18

Note: Summary statistics across all country-pairs and sectors in the world, reporting percentage decreases in trade costs due to the BRI. Estimates come from De Soyres et al. (2019)

Table A.4: Results by Scenario for the Containerized Goods Sector (%)

Country	**Lower**			**Central**			**Upper**		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
ABW	−0.01	−0.02	−0.01	−0.01	−0.02	−0.04	−0.02	−0.04	−0.06
AFG	−0.03	−0.02	0	−0.12	−0.05	0	−0.11	−0.11	−0.01
AGO	−0.02	−0.03	0	−0.08	−0.1	0	−0.14	−0.22	0
AIA	−0.01	−0.01	0	−0.04	0.01	−0.04	−0.07	0	−0.06
ALB	−0.02	−0.02	−0.01	0.03	0.09	−0.08	0.12	0.21	−0.18
AND	−0.01	−0.02	0	−0.05	−0.02	0	−0.1	−0.05	−0.01
ARE	0.52	10.81	−0.06	2.65	48.55	−3.16	5.25	84.83	−5.26
ARG	−0.01	−0.04	0	−0.03	−0.13	0	−0.06	−0.23	0
ARM	−0.03	−0.04	0	0.07	0.13	−0.06	0.21	0.27	−0.13
ASM	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.05	0
ATG	−0.01	−0.02	0	−0.06	−0.06	0	−0.1	−0.12	0
AUS	0	−0.08	0	−0.02	−0.31	0	−0.03	−0.41	0
AUT	0	−0.03	−0.01	0.02	0.17	−0.09	0.08	0.48	−0.23
AZE	−0.03	−0.06	0	0.22	0.44	−0.07	0.54	0.93	−0.15
BDI	−0.03	−0.02	0	−0.1	−0.06	0	−0.19	−0.14	0
BEL	0	−0.03	0	0	0.01	−0.08	0	0.04	−0.16
BEN	−0.02	−0.01	0	−0.01	−0.01	0	−0.01	−0.04	0
BES	−0.01	−0.01	0	−0.06	−0.02	0	−0.11	−0.04	0
BFA	−0.02	−0.02	0	−0.02	−0.01	−0.03	−0.03	−0.04	−0.04
BGD	−0.01	−0.05	0	0.04	0.32	−0.03	0.06	0.49	−0.04
BGR	0	0	−0.03	0.1	0.31	−0.13	0.28	0.73	−0.27

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
BHR	0.13	0.85	−0.02	2.43	12.12	−1.42	6.56	27.88	−3.06
BHS	−0.01	−0.03	0	0.08	0.32	−0.08	0.15	0.58	−0.13
BIH	0	0	−0.02	0.11	0.21	−0.11	0.36	0.57	−0.27
BLR	−0.01	−0.03	0	−0.01	0.05	−0.07	−0.03	0.03	−0.12
BLZ	−0.01	−0.01	0	0.01	0.01	−0.03	0.05	0.02	−0.06
BMU	−0.01	−0.01	0	−0.05	−0.05	0	−0.09	−0.09	0
BOL	−0.01	−0.02	0	−0.05	−0.06	0	−0.08	−0.1	−0.01
BRA	0	−0.06	−0.01	−0.02	−0.26	0	−0.04	−0.46	0
BRB	−0.01	−0.02	0	0.1	0.24	−0.05	0.21	0.47	−0.07
BRN	−0.01	−0.02	0	−0.03	−0.03	−0.01	−0.06	−0.08	−0.01
BTN	−0.02	−0.01	0	−0.02	0.04	−0.01	−0.05	0.02	−0.02
BWA	−0.02	−0.02	0	−0.07	−0.06	0	−0.13	−0.14	0
CAF	−0.03	−0.02	0	−0.1	−0.03	0	−0.12	−0.05	−0.01
CAN	0	−0.02	−0.01	0.02	0.08	−0.1	0.03	0.16	−0.19
CCK	−0.02	−0.01	0	−0.08	−0.02	0	−0.14	−0.07	0
CHE	0	−0.04	0	−0.01	−0.13	−0.01	−0.03	−0.23	−0.05
CHL	−0.01	−0.05	0	−0.03	−0.2	0	−0.05	−0.33	0
CHN	0	−0.06	0	0	−0.17	0	−0.01	−0.26	−0.01
CIV	−0.01	−0.01	−0.01	0.05	0.06	−0.07	0.1	0.1	−0.14
CMR	−0.02	−0.02	0	−0.08	−0.06	0	−0.11	−0.09	−0.03
COD	−0.02	−0.02	0	−0.08	−0.05	0	−0.14	−0.12	0
COG	−0.01	−0.01	0	−0.06	−0.02	0	−0.11	−0.07	0
COK	−0.01	−0.02	0	−0.06	−0.04	0	−0.1	−0.08	0
COL	0	−0.03	−0.01	0.02	0.15	−0.1	0.05	0.3	−0.19
COM	−0.03	−0.02	0	−0.11	−0.05	0	−0.19	−0.12	0
CPV	−0.02	−0.02	0	0.02	0.05	−0.04	0.05	0.05	−0.06
CRI	−0.01	−0.02	0	0.04	0.19	−0.08	0.09	0.35	−0.14
CUB	−0.01	−0.01	0	0.05	0.1	−0.12	0.08	0.16	−0.19
CUW	−0.01	−0.02	0	−0.05	−0.06	0	−0.1	−0.11	0
CXR	−0.02	−0.01	0	−0.07	−0.02	0	−0.12	−0.06	0
CYM	0	−0.05	0	−0.01	−0.16	0	−0.02	−0.24	0
CYP	2.93	6.16	−0.14	5.27	10.62	−0.39	9.21	16.49	−0.69
CZE	0	−0.03	0	0.01	0.1	−0.08	0.02	0.16	−0.16
DEU	0	−0.04	−0.01	0.02	0.31	−0.11	0.03	0.59	−0.21

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
DJI	−0.04	−0.07	0	−0.15	−0.22	0	−0.26	−0.44	0
DMA	−0.01	−0.01	0	0.04	0.08	−0.08	0.09	0.12	−0.12
DNK	0	−0.02	−0.01	0.02	0.18	−0.11	0.04	0.31	−0.22
DOM	−0.01	−0.03	0	0.06	0.27	−0.08	0.13	0.53	−0.14
DZA	0	0	−0.03	0.19	0.37	−0.15	0.67	1.1	−0.3
ECU	−0.01	−0.03	0	0.04	0.19	−0.12	0.08	0.36	−0.2
EGY	0.07	0.3	−0.05	0.24	0.98	−0.13	1.01	3.63	−0.28
ERI	−0.05	−0.03	0	−0.17	−0.08	0	−0.3	−0.19	0
ESH	−0.02	−0.01	0	−0.08	−0.01	0	−0.15	−0.04	0
ESP	0	−0.05	−0.01	0.02	0.33	−0.09	0.07	0.89	−0.21
EST	−0.01	−0.02	0	−0.02	0.03	−0.06	−0.03	0.01	−0.11
ETH	−0.04	−0.05	0	−0.13	−0.17	0	−0.23	−0.36	0
FIN	−0.01	−0.03	0	0.04	0.21	−0.1	0.08	0.35	−0.18
FJI	−0.01	−0.02	0	−0.02	0	−0.02	−0.04	−0.03	−0.03
FLK	−0.02	−0.01	0	−0.07	−0.02	0	−0.12	−0.05	0
FRA	0	−0.04	0	0	0.01	−0.04	0.01	0.17	−0.12
FRO	−0.01	−0.03	0	−0.05	−0.08	0	−0.1	−0.17	0
FSM	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.05	0
GAB	−0.01	−0.06	0	−0.04	−0.25	0	−0.07	−0.47	0
GBR	0	−0.06	−0.01	0.02	0.39	−0.11	0.04	0.85	−0.22
GEO	−0.03	−0.04	0	0.15	0.32	−0.1	0.39	0.69	−0.22
GHA	0	−0.05	−0.02	0.02	0.3	−0.06	0.03	0.5	−0.1
GIB	−0.01	−0.06	0	−0.05	−0.22	0	−0.09	−0.45	0
GIN	−0.01	−0.02	0	0.08	0.11	−0.05	0.17	0.18	−0.09
GMB	−0.02	−0.01	0	0.02	0	−0.01	0.06	−0.02	−0.01
GNB	−0.02	−0.01	0	0.01	−0.01	0	0.04	−0.05	0
GNQ	−0.02	−0.02	0	−0.08	−0.03	0	−0.14	−0.08	0
GRC	−0.01	−0.02	−0.02	0.09	0.44	−0.1	0.41	1.62	−0.25
GRD	−0.01	−0.02	0	−0.01	−0.02	0	−0.01	−0.04	0
GRL	−0.01	−0.01	0	−0.07	−0.02	0	−0.12	−0.06	0
GTM	−0.01	−0.02	−0.01	0.08	0.28	−0.15	0.14	0.47	−0.25
GUM	−0.01	−0.02	0	−0.05	−0.04	0	−0.09	−0.08	0
GUY	−0.01	−0.01	0	0.11	0.18	−0.1	0.25	0.35	−0.16
HKG	0	−0.07	0	0	−0.23	0	0	−0.34	0

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
HND	0	−0.01	−0.01	0.07	0.14	−0.11	0.14	0.24	−0.17
HRV	−0.01	−0.02	−0.01	0.08	0.23	−0.11	0.28	0.67	−0.25
HTI	−0.01	−0.03	0	0.04	0.21	−0.09	0.07	0.35	−0.14
HUN	0	−0.02	−0.02	0.01	0.1	−0.07	0.03	0.23	−0.17
IDN	0	−0.04	0	0.06	0.51	−0.03	0.11	0.95	−0.05
IND	0.03	0.43	−0.03	0.12	1.68	−0.06	0.16	2.11	−0.2
IOT	−0.03	−0.01	0	−0.1	−0.03	0	−0.18	−0.1	0
IRL	0	−0.02	−0.01	0.03	0.22	−0.12	0.07	0.44	−0.24
IRN	−0.03	−0.12	0	0.29	1.25	−0.25	0.97	3.78	−0.53
IRQ	−0.07	−0.17	0	0.39	1.07	−0.06	1.06	2.51	−0.11
ISL	−0.01	−0.02	−0.01	0.1	0.18	−0.12	0.23	0.37	−0.23
ISR	0	−0.02	−0.01	0.16	1.55	−0.14	0.54	4.41	−0.3
ITA	0	−0.06	0	0	0.02	−0.03	0.02	0.35	−0.13
JAM	−0.01	−0.02	0	0.09	0.22	−0.1	0.18	0.39	−0.16
JOR	3.53	7.46	−0.9	4.92	10.34	−1.06	7.21	13.97	−1.37
JPN	0	−0.04	0	0	−0.03	−0.02	0	−0.05	−0.03
KAZ	−0.02	−0.03	0	−0.04	−0.02	−0.04	−0.05	−0.05	−0.09
KEN	−0.03	−0.05	0	−0.1	−0.15	0	−0.18	−0.32	0
KGZ	−0.02	−0.02	0	−0.09	−0.05	0	−0.09	−0.08	−0.03
KHM	0	−0.02	0	−0.01	−0.03	−0.01	−0.03	−0.07	−0.02
KIR	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.05	0
KNA	−0.01	−0.01	0	0.03	0.07	−0.03	0.06	0.1	−0.05
KOR	0	−0.03	0	0	−0.07	−0.01	−0.01	−0.12	−0.01
KWT	0.38	2.19	−0.03	3.4	15.45	−1.03	7.22	27.87	−2.01
LAO	−0.01	−0.01	0	−0.04	−0.03	0	−0.08	−0.07	0
LBN	0.37	0.99	−0.05	0.55	1.58	−0.13	0.88	2.41	−0.28
LBR	−0.02	−0.04	0	0.03	0.07	−0.02	0.05	0.1	−0.03
LBY	0.02	0.02	−0.04	0.1	0.12	−0.07	0.29	0.23	−0.1
LCA	−0.01	−0.01	0	0.04	0.08	−0.03	0.09	0.12	−0.05
LKA	−0.01	−0.06	0	−0.03	−0.14	−0.01	−0.06	−0.31	−0.01
LSO	−0.01	−0.04	0	−0.05	−0.13	0	−0.1	−0.26	0
LTU	−0.01	−0.03	−0.01	0.04	0.15	−0.1	0.07	0.23	−0.19
LUX	−0.01	−0.02	0	−0.04	−0.06	0	−0.08	−0.13	0
LVA	−0.01	−0.02	0	0.03	0.11	−0.09	0.05	0.16	−0.16

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
MAC	0	−0.04	0	0	−0.08	−0.01	−0.01	−0.13	−0.02
MAR	0	0	−0.03	0.09	0.39	−0.13	0.29	1.14	−0.28
MDA	−0.01	−0.02	−0.01	0.08	0.17	−0.11	0.21	0.36	−0.22
MDG	−0.02	−0.02	0	−0.1	−0.07	0	−0.17	−0.15	0
MDV	−0.02	−0.07	0	−0.08	−0.25	0	−0.13	−0.43	0
MEX	0	−0.03	−0.01	0.01	0.21	−0.1	0.02	0.34	−0.18
MHL	0	−0.1	0	−0.02	−0.39	0	−0.03	−0.68	0
MKD	0	0	−0.03	0.07	0.21	−0.11	0.21	0.52	−0.25
MLI	−0.02	−0.01	0	0.06	0.11	−0.06	0.1	0.15	−0.1
MLT	0	−0.02	−0.01	0.08	0.45	−0.09	0.44	2.22	−0.27
MMR	−0.01	−0.01	0	−0.01	0.04	−0.02	−0.02	0.03	−0.03
MNE	−0.01	−0.01	−0.01	0.08	0.11	−0.04	0.31	0.28	−0.09
MNG	−0.01	−0.01	0	−0.04	−0.02	0	−0.08	−0.05	0
MNP	−0.01	−0.01	0	−0.05	−0.02	0	−0.1	−0.05	0
MOZ	−0.02	−0.02	0	−0.08	−0.06	0	−0.14	−0.14	0
MRT	−0.02	−0.01	0	0.07	0.07	−0.06	0.17	0.11	−0.11
MSR	−0.01	−0.01	0	−0.05	−0.01	0	−0.08	−0.03	0
MUS	−0.02	−0.06	0	−0.07	−0.22	0	−0.12	−0.43	0
MWI	−0.02	−0.03	0	−0.09	−0.09	0	−0.16	−0.19	0
MYS	0	−0.04	0	−0.01	−0.08	−0.01	−0.02	−0.17	−0.01
NAM	−0.02	−0.03	0	−0.08	−0.07	0	−0.14	−0.16	0
NCL	−0.01	−0.02	0	−0.05	−0.06	0	−0.1	−0.12	0
NER	−0.02	−0.02	0	0.05	0.09	−0.03	0.07	0.11	−0.04
NFK	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.04	0
NGA	−0.01	−0.07	−0.01	−0.03	−0.24	−0.01	−0.06	−0.43	−0.02
NIC	−0.01	−0.02	0	0.08	0.19	−0.09	0.17	0.35	−0.17
NIU	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.04	0
NLD	0	−0.04	0	0.02	0.28	−0.11	0.04	0.57	−0.22
NOR	0	−0.02	−0.01	0.04	0.19	−0.09	0.08	0.32	−0.17
NPL	−0.02	−0.02	0	−0.07	−0.03	0	−0.12	−0.11	0
NRU	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.05	0
NZL	−0.01	−0.05	0	−0.01	−0.13	−0.01	−0.03	−0.27	−0.01
OMN	0.62	1.51	−0.03	1.06	2.46	−0.06	3.12	6.56	−1
PAK	0.08	0.28	−0.02	0.1	0.37	−0.03	0.36	1.17	−0.44

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
PAN	−0.01	−0.04	0	0.04	0.29	−0.08	0.1	0.68	−0.14
PCN	−0.01	−0.01	0	−0.06	−0.01	0	−0.1	−0.03	0
PER	0	−0.03	−0.01	0.02	0.12	−0.07	0.03	0.19	−0.12
PHL	0	−0.04	0	−0.01	−0.08	−0.01	−0.03	−0.15	−0.01
PLW	−0.01	−0.01	0	−0.05	−0.02	0	−0.1	−0.04	0
PNG	−0.01	−0.01	0	−0.05	−0.03	0	−0.1	−0.07	0
POL	0	−0.05	0	0.02	0.19	−0.08	0.03	0.33	−0.16
PRK	−0.01	−0.01	0	−0.03	−0.02	0	−0.05	−0.03	0
PRT	0	−0.02	−0.02	0.01	0.12	−0.08	0.03	0.27	−0.18
PRY	−0.01	−0.02	0	−0.04	−0.08	0	−0.08	−0.15	0
PSE	0.5	1.13	−0.06	0.72	1.61	−0.08	1.02	1.95	−0.11
PYF	−0.01	−0.03	0	−0.04	−0.11	0	−0.08	−0.22	0
QAT	0.24	1.3	−0.01	2.51	12.5	−0.97	6.43	27.56	−2.04
ROU	0	−0.02	−0.02	0.09	0.53	−0.14	0.26	1.38	−0.3
RUS	−0.01	−0.03	−0.01	0.07	0.33	−0.14	0.16	0.72	−0.27
RWA	−0.03	−0.04	0	−0.1	−0.11	0	−0.18	−0.24	0
SAU	1.51	8.41	−0.17	3.04	15.68	−0.4	4.31	20.64	−0.74
SDN	−0.04	−0.04	0	−0.14	−0.13	0	−0.24	−0.28	0
SEN	−0.01	−0.02	0	0.08	0.14	−0.1	0.2	0.3	−0.21
SGP	0	−0.1	0	0	−0.2	0	−0.01	−0.27	0
SHN	−0.02	−0.01	0	−0.08	−0.02	0	−0.15	−0.07	0
SLB	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.05	0
SLE	−0.02	−0.01	0	0.05	0.05	−0.07	0.11	0.06	−0.12
SLV	0	−0.02	−0.01	0.05	0.2	−0.09	0.1	0.38	−0.15
SMR	−0.01	−0.01	0	−0.06	−0.01	0	−0.13	−0.04	0
SOM	−0.04	−0.03	0	−0.13	−0.08	0	−0.23	−0.19	0
SPM	−0.01	−0.01	0	−0.06	−0.01	0	−0.12	−0.03	0
SRB	0	−0.01	−0.02	0.11	0.24	−0.12	0.36	0.66	−0.27
SSD	−0.03	−0.02	0	−0.12	−0.05	0	−0.22	−0.13	0
STP	−0.02	−0.02	0	−0.08	−0.03	0	−0.13	−0.08	0
SUR	−0.01	−0.01	0	0.05	0.04	−0.07	0.17	0.1	−0.11
SVK	0	−0.02	0	0.04	0.22	−0.1	0.15	0.63	−0.26
SVN	0	−0.01	−0.01	0.03	0.13	−0.08	0.1	0.37	−0.21
SWE	−0.01	−0.04	−0.01	0.04	0.31	−0.11	0.09	0.55	−0.2

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
SWZ	−0.02	−0.02	0	−0.06	−0.06	0	−0.11	−0.14	0
SXM	−0.01	−0.03	0	−0.05	−0.14	0	−0.09	−0.27	0
SYC	−0.02	−0.09	0	−0.06	−0.34	0	−0.11	−0.64	0
SYR	−0.09	−0.07	0	−0.21	−0.13	0	−0.37	−0.3	0
TCA	−0.01	−0.01	0	−0.03	−0.02	0	−0.05	−0.03	0
TCD	−0.03	−0.02	0	−0.1	−0.03	0	−0.13	−0.04	−0.02
TGO	−0.02	−0.01	0	0.04	0.07	−0.02	0.07	0.09	−0.04
THA	0	−0.04	0	−0.01	−0.07	−0.01	−0.02	−0.16	−0.01
TJK	−0.03	−0.02	0	−0.11	−0.05	0	−0.12	−0.06	−0.04
TKL	−0.01	−0.01	0	−0.06	−0.01	0	−0.1	−0.04	0
TKM	−0.04	−0.02	0	0.76	0.7	−0.14	1.83	1.57	−0.31
TLS	−0.01	−0.01	0	−0.06	−0.02	0	−0.11	−0.06	0
TON	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.06	0
TTO	−0.01	−0.03	0	0.08	0.29	−0.07	0.18	0.57	−0.12
TUN	0	0	−0.03	0.08	0.19	−0.09	0.48	0.82	−0.25
TUR	−0.01	−0.07	0	0.07	0.64	−0.11	0.24	1.9	−0.28
TUV	−0.01	−0.01	0	−0.06	−0.03	0	−0.1	−0.06	0
TWN	0	−0.02	0	−0.01	−0.05	−0.01	−0.02	−0.09	−0.01
TZA	−0.02	−0.05	0	−0.09	−0.15	0	−0.16	−0.31	0
UGA	−0.03	−0.04	0	−0.1	−0.11	0	−0.19	−0.24	0
UKR	0	0.01	−0.03	0.17	0.56	−0.18	0.55	1.59	−0.36
URY	−0.01	−0.03	0	−0.03	−0.09	0	−0.06	−0.17	0
USA	0	−0.07	−0.01	0.02	0.82	−0.1	0.02	1.29	−0.17
UZB	0.06	0.19	−0.02	0.05	0.26	−0.04	0.07	0.27	−0.08
VCT	−0.01	−0.01	0	−0.03	−0.02	0	−0.03	−0.04	0
VEN	−0.01	−0.01	0	0.1	0.13	−0.08	0.21	0.22	−0.13
VGB	−0.01	−0.04	0	−0.02	−0.15	0	−0.04	−0.23	0
VNM	0	−0.05	0	0	−0.11	−0.01	−0.01	−0.18	−0.01
VUT	−0.01	−0.01	0	−0.06	−0.02	0	−0.1	−0.05	0
WLF	−0.01	−0.01	0	−0.06	−0.02	0	−0.11	−0.05	0
WSM	−0.01	−0.01	0	−0.06	−0.03	0	−0.1	−0.06	0
YEM	−0.08	−0.06	0	−0.23	−0.16	0	−0.38	−0.33	0
ZAF	−0.01	−0.05	0	−0.04	−0.17	0	−0.07	−0.33	0
ZMB	−0.02	−0.02	0	−0.08	−0.06	0	−0.14	−0.14	0

Continued on next page

Table A.4 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs	RGDP	Exports	Trade Costs
ZWE	−0.02	−0.03	0	−0.08	−0.09	0	−0.14	−0.19	0

Note: All variables capture percentage changes in the containerized goods sector only, not aggregates.

Table A.5: Results in Level-Changes by Scenario

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
ABW	0.0	0.0	−0.01	0.0	0.0	−0.04	0.0	−0.1	−0.06
AFG	−0.4	−0.2	0	−1.5	−0.6	0	−1.4	−1.3	−0.01
AGO	−0.6	−0.9	0	−2.4	−2.9	0	−4.4	−6.0	0
AIA	0.0	0.0	0	0.0	0.0	−0.04	0.0	0.0	−0.06
ALB	−0.5	−0.5	−0.01	0.9	2.9	−0.08	4.0	6.5	−0.18
AND	0.0	0.0	0	0.0	0.0	0	−0.1	0.0	−0.01
ARE	888.4	5339.1	−0.06	4531.8	23 984.2	−3.16	8979.2	41 905.0	−5.26
ARG	−2.9	−10.4	0	−13.8	−38.2	0	−25.6	−69.4	0
ARM	−0.8	−0.9	0	1.9	3.2	−0.06	5.7	6.6	−0.13
ASM	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
ATG	0.0	0.0	0	−0.1	−0.1	0	−0.2	−0.2	0
AUS	−9.7	−60.6	0	−41.8	−232.7	0	−60.5	−305.9	0
AUT	−6.6	−29.6	−0.01	42.2	186.9	−0.09	143.5	519.6	−0.23
AZE	−1.2	−1.8	0	8.4	14.3	−0.07	20.6	30.2	−0.15
BDI	−0.1	−0.1	0	−0.5	−0.3	0	−1.0	−0.7	0
BEL	−8.4	−44.5	0	−9.5	17.7	−0.08	−14.0	52.0	−0.16
BEN	−0.1	−0.1	0	0.0	−0.1	0	−0.1	−0.2	0
BES	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
BFA	−0.5	−0.4	0	−0.5	−0.2	−0.03	−0.9	−0.9	−0.04
BGD	−9.9	−32.9	0	41.1	201.6	−0.03	69.3	302.4	−0.04
BGR	−0.3	0.5	−0.03	32.9	80.8	−0.13	94.0	191.6	−0.27
BHR	8.5	34.9	−0.02	159.5	497.3	−1.42	431.2	1144.1	−3.06
BHS	−0.2	−0.6	0	2.3	6.9	−0.08	4.4	12.7	−0.13
BIH	0.0	−0.2	−0.02	6.9	11.5	−0.11	22.7	30.9	−0.27

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
BLR	−2.9	−5.1	0	−2.6	9.0	−0.07	−7.4	4.8	−0.12
BLZ	0.0	0.0	0	0.0	0.0	−0.03	0.1	0.0	−0.06
BMU	−0.1	−0.2	0	−0.7	−0.6	0	−1.3	−1.1	0
BOL	−0.6	−1.1	0	−3.2	−3.7	0	−5.2	−6.1	−0.01
BRA	−10.4	−61.2	−0.01	−56.4	−282.6	0	−107.5	−499.1	0
BRB	0.0	−0.1	0	0.3	0.6	−0.05	0.6	1.2	−0.07
BRN	0.0	0.0	0	−0.1	−0.1	−0.01	−0.2	−0.2	−0.01
BTN	0.0	0.0	0	0.0	0.0	−0.01	0.0	0.0	−0.02
BWA	−0.5	−0.6	0	−2.0	−1.7	0	−3.7	−3.8	0
CAF	0.0	0.0	0	0.0	0.0	0	0.0	0.0	−0.01
CAN	−9.6	−37.5	−0.01	63.7	181.8	−0.1	112.1	368.7	−0.19
CCK	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
CHE	−13.3	−77.4	0	−60.7	−225.0	−0.01	−119.2	−402.2	−0.05
CHL	−2.3	−11.4	0	−12.0	−49.1	0	−21.2	−81.3	0
CHN	−125.6	−1089.7	0	−467.1	−3259.4	0	−1099.7	−4927.8	−0.01
CIV	−0.4	−0.4	−0.01	1.4	1.8	−0.07	3.2	2.9	−0.14
CMR	−0.5	−0.6	0	−2.3	−1.7	0	−3.1	−2.4	−0.03
COD	−1.9	−1.9	0	−7.1	−4.5	0	−12.6	−10.4	0
COG	−0.4	−0.4	0	−1.9	−0.8	0	−3.6	−2.2	0
COK	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
COL	−2.2	−9.4	−0.01	10.4	42.4	−0.1	21.7	86.7	−0.19
COM	0.0	0.0	0	0.0	0.0	0	−0.1	0.0	0
CPV	0.0	0.0	0	0.0	0.1	−0.04	0.1	0.1	−0.06
CRI	−0.6	−2.1	0	5.1	16.1	−0.08	10.5	30.2	−0.14
CUB	−0.1	−0.1	0	0.5	0.9	−0.12	0.8	1.5	−0.19
CUW	0.0	0.0	0	−0.1	−0.1	0	−0.2	−0.2	0
CXR	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
CYM	−0.5	−3.1	0	−1.7	−10.3	0	−3.3	−15.2	0
CYP	115.1	195.4	−0.14	207.4	336.8	−0.39	362.4	522.8	−0.69
CZE	−8.1	−36.5	0	22.9	116.8	−0.08	41.9	192.6	−0.16
DEU	−36.0	−255.1	−0.01	345.5	2021.0	−0.11	705.3	3860.2	−0.21
DJI	−0.3	−0.4	0	−1.0	−1.3	0	−1.7	−2.6	0
DMA	0.0	0.0	0	0.0	0.0	−0.08	0.0	0.0	−0.12

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
DNK	−3.7	−15.7	−0.01	20.1	114.4	−0.11	41.7	202.6	−0.22
DOM	−0.7	−2.3	0	6.7	21.6	−0.08	14.0	41.6	−0.14
DZA	−0.1	0.0	−0.03	8.2	13.8	−0.15	28.6	41.4	−0.3
ECU	−0.8	−2.7	0	5.1	16.5	−0.12	10.4	31.8	−0.2
EGY	34.3	104.7	−0.05	120.0	345.0	−0.13	500.3	1275.2	−0.28
ERI	−0.1	−0.1	0	−0.4	−0.2	0	−0.7	−0.5	0
ESH	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
ESP	−14.4	−80.1	−0.01	99.8	547.8	−0.09	301.2	1498.0	−0.21
EST	−1.1	−1.9	0	−1.8	2.5	−0.06	−3.6	1.2	−0.11
ETH	−1.7	−2.2	0	−5.9	−7.0	0	−10.5	−14.6	0
FIN	−6.4	−18.8	0	34.9	126.7	−0.1	70.5	217.3	−0.18
FJI	−0.1	−0.2	0	−0.2	0.0	−0.02	−0.4	−0.3	−0.03
FLK	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
FRA	−26.3	−153.0	0	−21.7	24.9	−0.04	73.2	591.5	−0.12
FRO	−0.3	−0.6	0	−1.5	−1.8	0	−2.8	−3.8	0
FSM	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
GAB	−5.3	−22.6	0	−24.6	−88.4	0	−46.0	−167.6	0
GBR	−15.0	−104.0	−0.01	114.0	715.2	−0.11	256.2	1569.6	−0.22
GEO	−0.8	−1.0	0	4.1	8.1	−0.1	11.0	17.4	−0.22
GHA	−7.6	−44.1	−0.02	43.4	258.0	−0.06	75.1	426.8	−0.1
GIB	0.0	−0.1	0	−0.1	−0.2	0	−0.1	−0.4	0
GIN	−0.3	−0.5	0	2.4	3.1	−0.05	4.9	4.9	−0.09
GMB	0.0	0.0	0	0.0	0.0	−0.01	0.0	0.0	−0.01
GNB	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
GNQ	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
GRC	−2.0	−4.2	−0.02	30.8	99.8	−0.1	134.8	366.1	−0.25
GRD	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
GRL	−0.2	−0.2	0	−0.9	−0.3	0	−1.6	−0.7	0
GTM	−0.3	−0.9	−0.01	5.3	14.9	−0.15	9.4	25.2	−0.25
GUM	0.0	0.0	0	0.0	0.0	0	−0.1	−0.1	0
GUY	−0.1	−0.2	0	1.3	2.1	−0.1	3.0	4.0	−0.16
HKG	−9.1	−80.9	0	−35.6	−265.1	0	−82.0	−394.4	0
HND	−0.3	−0.6	−0.01	4.3	8.1	−0.11	8.4	13.8	−0.17

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
HRV	−1.3	−2.4	−0.01	11.7	27.1	−0.11	41.6	78.9	−0.25
HTI	−1.0	−3.6	0	6.3	23.7	−0.09	11.6	40.2	−0.14
HUN	−4.1	−15.8	−0.02	10.1	80.6	−0.07	42.6	189.8	−0.17
IDN	−9.8	−48.3	0	141.6	607.7	−0.03	272.0	1126.0	−0.05
IND	239.8	1352.7	−0.03	935.0	5258.2	−0.06	1286.2	6621.0	−0.2
IOT	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
IRL	−6.2	−24.3	−0.01	56.9	252.4	−0.12	131.9	509.8	−0.24
IRN	−26.3	−71.7	0	260.7	758.9	−0.25	870.4	2297.2	−0.53
IRQ	−3.4	−7.0	0	20.0	42.9	−0.06	54.8	101.1	−0.11
ISL	−0.5	−0.7	−0.01	4.6	7.9	−0.12	11.1	16.1	−0.23
ISR	−0.9	−5.1	−0.01	85.1	405.0	−0.14	276.3	1154.0	−0.3
ITA	−31.1	−182.3	0	−7.9	52.6	−0.03	202.7	1102.3	−0.13
JAM	0.0	0.0	0	0.2	0.4	−0.1	0.4	0.7	−0.16
JOR	467.0	795.5	−0.9	651.5	1101.8	−1.06	954.2	1489.6	−1.37
JPN	−20.2	−182.6	0	−29.5	−144.3	−0.02	−99.1	−240.7	−0.03
KAZ	−4.0	−5.6	0	−7.6	−3.6	−0.04	−9.9	−9.5	−0.09
KEN	−1.3	−1.9	0	−4.6	−6.2	0	−8.2	−12.8	0
KGZ	−0.7	−0.5	0	−2.7	−1.4	0	−2.8	−2.2	−0.03
KHM	−4.1	−12.0	0	−12.0	−17.3	−0.01	−24.2	−40.7	−0.02
KIR	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
KNA	0.0	0.0	0	0.0	0.0	−0.03	0.0	0.1	−0.05
KOR	−13.5	−102.6	0	−41.4	−208.7	−0.01	−97.7	−344.6	−0.01
KWT	27.4	100.6	−0.03	246.9	708.0	−1.03	524.3	1277.1	−2.01
LAO	−1.0	−1.3	0	−4.4	−2.7	0	−7.8	−6.3	0
LBN	21.9	44.5	−0.05	32.7	71.0	−0.13	52.4	108.2	−0.28
LBR	−0.2	−0.3	0	0.3	0.6	−0.02	0.6	0.9	−0.03
LBY	0.2	0.3	−0.04	1.3	1.6	−0.07	4.1	3.1	−0.1
LCA	0.0	0.0	0	0.0	0.0	−0.03	0.0	0.1	−0.05
LKA	−3.5	−11.3	0	−9.6	−26.5	−0.01	−18.1	−57.9	−0.01
LSO	−0.8	−2.0	0	−3.3	−6.4	0	−6.0	−12.7	0
LTU	−2.1	−4.3	−0.01	7.5	24.4	−0.1	14.1	38.9	−0.19
LUX	−0.8	−2.0	0	−4.4	−5.0	0	−8.6	−10.6	0
LVA	−1.4	−2.4	0	3.5	11.2	−0.09	5.9	16.0	−0.16

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
MAC	−1.2	−9.9	0	−3.7	−20.7	−0.01	−9.0	−33.7	−0.02
MAR	0.0	0.8	−0.03	36.5	113.1	−0.13	120.1	327.0	−0.28
MDA	−0.4	−0.4	−0.01	1.9	3.9	−0.11	5.3	8.1	−0.22
MDG	−0.6	−0.5	0	−2.2	−1.5	0	−4.0	−3.5	0
MDV	−0.1	−0.3	0	−0.6	−1.3	0	−0.9	−2.2	0
MEX	−10.4	−76.0	−0.01	91.9	547.5	−0.1	140.9	875.0	−0.18
MHL	0.0	−0.3	0	−0.2	−1.1	0	−0.3	−1.9	0
MKD	−0.2	0.3	−0.03	7.6	20.1	−0.11	24.4	49.3	−0.25
MLI	−1.1	−0.9	0	3.6	6.6	−0.06	6.5	9.7	−0.1
MLT	−0.1	−0.5	−0.01	3.2	11.7	−0.09	17.9	57.5	−0.27
MMR	−1.0	−0.9	0	−1.0	3.6	−0.02	−1.8	3.0	−0.03
MNE	−0.1	0.0	−0.01	0.3	0.4	−0.04	1.2	1.0	−0.09
MNG	−0.2	−0.2	0	−0.8	−0.3	0	−1.5	−0.9	0
MNP	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
MOZ	−0.5	−0.6	0	−2.0	−1.6	0	−3.6	−3.7	0
MRT	−0.1	−0.1	0	0.4	0.4	−0.06	1.1	0.7	−0.11
MSR	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
MUS	−0.3	−0.9	0	−1.3	−3.3	0	−2.4	−6.4	0
MWI	−0.3	−0.4	0	−1.3	−1.2	0	−2.3	−2.6	0
MYS	−8.2	−38.0	0	−21.3	−75.2	−0.01	−44.1	−158.3	−0.01
NAM	−0.4	−0.5	0	−1.7	−1.5	0	−3.1	−3.3	0
NCL	0.0	−0.1	0	−0.2	−0.2	0	−0.3	−0.4	0
NER	−0.1	−0.1	0	0.2	0.5	−0.03	0.4	0.6	−0.04
NFK	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
NGA	−8.4	−38.0	−0.01	−31.3	−126.1	−0.01	−56.6	−228.2	−0.02
NIC	−0.4	−0.7	0	4.2	8.9	−0.09	8.9	16.5	−0.17
NIU	0.0	0.0	0	0.0	0.0	0	−0.1	0.0	0
NLD	−9.8	−67.6	0	81.0	488.9	−0.11	174.9	982.6	−0.22
NOR	−3.0	−9.2	−0.01	24.6	82.1	−0.09	46.8	137.5	−0.17
NPL	−0.6	−0.5	0	−2.2	−0.9	0	−3.8	−3.1	0
NRU	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
NZL	−1.1	−6.1	0	−3.3	−15.1	−0.01	−7.0	−29.9	−0.01
OMN	57.9	113.1	−0.03	98.3	183.7	−0.06	290.6	490.1	−1

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
PAK	63.0	155.6	−0.02	80.8	208.9	−0.03	278.5	656.0	−0.44
PAN	−0.4	−1.7	0	2.8	12.3	−0.08	7.0	29.0	−0.14
PCN	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
PER	−2.2	−9.7	−0.01	9.9	38.0	−0.07	14.8	59.0	−0.12
PHL	−4.5	−19.3	0	−12.8	−41.1	−0.01	−26.1	−84.8	−0.01
PLW	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
PNG	−0.4	−0.4	0	−1.7	−0.9	0	−3.1	−2.2	0
POL	−19.9	−102.2	0	81.5	419.4	−0.08	155.5	724.8	−0.16
PRK	−0.6	−1.1	0	−2.9	−1.5	0	−5.2	−3.1	0
PRT	−2.1	−9.0	−0.02	7.8	52.3	−0.08	24.3	122.0	−0.18
PRY	−0.5	−1.0	0	−2.2	−3.5	0	−4.1	−6.5	0
PSE	13.0	24.8	−0.06	18.7	35.2	−0.08	26.7	42.6	−0.11
PYF	0.0	−0.1	0	−0.2	−0.5	0	−0.4	−0.9	0
QAT	22.3	72.9	−0.01	228.2	702.1	−0.97	584.9	1547.5	−2.04
ROU	−4.4	−12.1	−0.02	94.0	343.0	−0.14	278.6	899.6	−0.3
RUS	−13.9	−41.4	−0.01	132.5	451.1	−0.14	314.7	974.3	−0.27
RWA	−0.4	−0.5	0	−1.4	−1.4	0	−2.4	−3.0	0
SAU	1435.5	4783.4	−0.17	2891.0	8915.3	−0.4	4102.2	11 736.5	−0.74
SDN	−2.5	−2.6	0	−8.5	−7.4	0	−15.2	−16.4	0
SEN	−0.5	−0.8	0	3.0	4.7	−0.1	7.4	9.9	−0.21
SGP	−2.8	−22.2	0	−4.8	−42.9	0	−12.0	−59.7	0
SHN	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
SLB	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
SLE	0.0	0.0	0	0.0	0.0	−0.07	0.1	0.0	−0.12
SLV	−0.4	−1.3	−0.01	3.8	12.2	−0.09	8.2	23.5	−0.15
SMR	0.0	0.0	0	−0.1	0.0	0	−0.2	−0.1	0
SOM	−0.1	−0.1	0	−0.3	−0.2	0	−0.5	−0.4	0
SPM	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
SRB	−0.6	−0.6	−0.02	14.4	27.7	−0.12	47.9	74.5	−0.27
SSD	0.0	0.0	0	−0.1	0.0	0	−0.2	−0.1	0
STP	0.0	0.0	0	−0.1	0.0	0	−0.1	−0.1	0
SUR	−0.3	−0.3	0	1.3	1.0	−0.07	4.0	2.3	−0.11
SVK	−2.9	−13.2	0	45.5	151.3	−0.1	155.2	432.1	−0.26

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
SVN	−0.9	−3.6	−0.01	9.1	31.6	−0.08	35.6	91.5	−0.21
SWE	−10.5	−38.7	−0.01	78.7	334.1	−0.11	160.2	606.8	−0.2
SWZ	−0.6	−0.9	0	−2.4	−2.3	0	−4.3	−5.1	0
SXM	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
SYC	−0.6	−2.3	0	−2.4	−8.3	0	−4.4	−15.7	0
SYR	−0.2	−0.1	0	−0.4	−0.2	0	−0.7	−0.5	0
TCA	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
TCD	−0.1	0.0	0	−0.3	−0.1	0	−0.3	−0.1	−0.02
TGO	−0.1	0.0	0	0.1	0.2	−0.02	0.2	0.3	−0.04
THA	−14.8	−73.3	0	−39.1	−141.0	−0.01	−82.6	−301.3	−0.01
TJK	−0.3	−0.2	0	−1.1	−0.5	0	−1.2	−0.5	−0.04
TKL	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
TKM	−0.2	−0.1	0	3.9	3.5	−0.14	9.3	7.8	−0.31
TLS	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
TON	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
TTO	−0.2	−0.5	0	1.6	4.6	−0.07	3.5	9.1	−0.12
TUN	−0.5	−0.4	−0.03	10.7	21.2	−0.09	63.4	92.0	−0.25
TUR	−30.0	−123.4	0	236.6	1074.9	−0.11	793.8	3166.9	−0.28
TUV	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
TWN	−5.2	−26.4	0	−17.9	−52.8	−0.01	−37.4	−102.6	−0.01
TZA	−2.4	−4.1	0	−9.0	−13.1	0	−16.2	−27.0	0
UGA	−0.8	−0.9	0	−2.8	−2.7	0	−5.0	−5.9	0
UKR	0.1	1.5	−0.03	58.4	147.3	−0.18	186.2	417.0	−0.36
URY	−1.0	−3.0	0	−4.7	−10.0	0	−8.6	−18.2	0
USA	−42.0	−355.9	−0.01	620.5	3981.1	−0.1	846.3	6239.6	−0.17
UZB	12.7	34.1	−0.02	12.1	47.0	−0.04	15.1	49.4	−0.08
VCT	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
VEN	−0.1	−0.1	0	0.8	1.0	−0.08	1.7	1.8	−0.13
VGB	−0.2	−0.7	0	−0.7	−2.5	0	−1.1	−3.9	0
VNM	−23.5	−188.5	0	−65.4	−439.1	−0.01	−164.1	−697.2	−0.01
VUT	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
WLF	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0
WSM	0.0	0.0	0	0.0	0.0	0	0.0	0.0	0

Continued on next page

Table A.5 – *Continued from previous page*

Country	Lower			Central			Upper		
	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)	RGDP ($M)	Exports ($M)	Trade Costs (%)
YEM	−0.5	−0.3	0	−1.4	−1.0	0	−2.4	−2.0	0
ZAF	−8.2	−27.2	0	−32.1	−91.3	0	−58.1	−176.7	0
ZMB	−2.1	−2.1	0	−7.9	−5.5	0	−14.3	−13.0	0
ZWE	−0.9	−1.3	0	−3.6	−3.7	0	−6.5	−8.0	0

Note: Real GDP (RGDP) and exports are reported as counterfactual minus baseline changes in levels, in millions of 2019 U.S. dollars. Trade costs are reported as simple-average percentage changes in bilateral trade costs by exporter; negative values denote reductions in trade costs.



PUBLICATIONS

Alternative Trade Corridors of the GCC: The Case of IMEC

Working Paper WP/2026/237